Becker Meets Kyle: Inside Insider Trading*

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Abstract

How do illegal insider traders act on private information? We answer this question using a unique sample of illegal insider traders convicted by the Securities Exchange Commission (SEC). To shed light on the traders' investment strategies, we analyze, theoretically and empirically, the tradeoff between the risk of information becoming public (information risk), and the risk of being subject to enforcement (legal risk). Using a regression model and various shocks to enforcement risk, we find that a subset of investors respond to this tradeoff by splitting their trades more and trading less aggressively in the presence of greater legal risk and smaller information risk. At the same time, a large fraction of traders do not internalize such a tradeoff. The insiders manage their trades' size according to prevailing liquidity conditions, volatility of returns, noise trading activity, and the precision of the signal. Individual characteristics, such as investing expertise and age, also play an essential role. Insiders facing increasing legal risk concentrate more on information of higher value. Thus, insider trading enforcement may hamper stock price informativeness.

1 Introduction

Information is transmitted into prices through the actions of informed traders.[1] While some of the traders act legally on private information they obtain from research or privileged corporate positions, others act on private information in breach of a fiduciary duty. Theoretical research on illegal insider trading has largely focused on aggregate outcomes, such as economic efficiency (Ausubel (1990)), capital formation process (Manove (1989)), and optimal enforcement framework (DeMarzo et al. (1998)). A few empirical applications are mostly testing the impact of illegal insider trading on security prices (Meulbroek (1992)) and public signals (Kacperczyk and Pagnotta (2019)). Much less attention, both theoretically and empirically, has been devoted to understanding ex-ante incentives of traders who are subject to litigation risk. Understanding insiders' trading strategies is important because it has implications for the aggregate outcomes. In particular, it can help the regulators improve their detection technologies. It can also help characterize the process by which private signals are revealed to the market as a whole. The aim of this paper is to lay out a rational framework of insider trading activity by extending a trading model of Kyle (1985) to include legal penalties in the spirit of Becker (1968) with a goal to provide an empirical assessment of such strategies and eventually their impact on price discovery.

We motivate our empirical tests using a simple two-period model, with investors deciding when and how much to trade. Apart from a typical transaction cost friction, as in Kyle, we add two additional features that are more specific to our setting. First, traders face information risk, that is, the possibility that information becomes public, and thus redundant, at a stochastic period. Second, they face risk associated with possible legal enforcement. In addition, and more germane to models of asymmetric information, traders'

[1] For example, Grossman (1976); Grossman and Stigliz (1980); Glosten and Milgrom (1985); Kyle (1985); Easley and Hara (1987), among others, in stock markets and Biais and Hillion (1994); Easley et al. (1998); Wang (1993); Back (1993), among others, in option markets.

incentives depend on the information environment in which they trade (signal strength and precision, fundamental volatility, and amount of noise trading). In this framework, we show that investors face a tradeoff to execute their trades faster, when faced with a greater degree of information risk, vs. to act more conservatively when faced with a higher expected penalty associated with their actions. The equilibrium outcome is a rational response to the relative strengths of the two forces and the information environment in which trading takes place.

We empirically evaluate the predictions of our model using a hand-collected sample of insider trading cases prosecuted by the U.S. Securities Exchange Commission (SEC) that document in detail how certain individuals trade on nonpublic and material information. These cases involve a large number of trades in several hundred firms over the period 1995–2015; hence, they are representative of a fairly large universe of assets and market conditions. A key advantage of our setting is that we can observe not only trades but the dynamics of market signals, that is, when private information is communicated to the trader. Our ability to observe the arrival of private information directly is in stark contrast to prior literature, which typically infers the presence of informed trading indirectly, usually by either observing the trading behavior of financial professionals (e.g., institutional investors or large activist shareholders), or trading ahead of important information events (e.g., earnings announcements or mergers). We also observe the detailed information on legal consequences associated with these traders' actions.

At the outset, we evaluate the economic characteristics of our sample. First, we assess the strength of the information on which individuals in our sample trade by computing hypothetical stock returns (excluding dividends) that an investor would realize if he initiated his trade at the opening price of the day the insider first trades and closed his trade at the opening price of the day following the public information disclosure. We show that, on average, such returns exceed 40% for private signals with a positive sign and 20% for those with a negative sign. Both results are economically large, especially since they accrue over a relatively short period of seven days, on average. These figures could, in fact, underestimate the pre-fees profits of the informed traders, since 30% of the trades in our sample are executed using options, not stocks.

Second, we document the stylized characteristics of the traders. Our sample contains both individuals with a small amount of available capital, wealthy individual investors, and institutional investors, such as hedge funds (e.g., Galleon Group, SAC Capital). A substantial fraction of the individuals work in the finance industry or have top executive jobs; hence, they exemplify how skilled investors trade. Finally, we analyze the scope of informed volume. We show that, on days when insiders trade, their trades constitute more than 10% of the total volume for stocks and more than 30% for options. Overall, our sample reflects the trades of an economically relevant group.

We next take a closer look at the insiders' trading strategies. Since trading strategies can be complex, in the interest of parsimony, we summarize the main strategic dimensions using two different metrics. First, we examine traders' incentives to split their trades over multiples dates (e.g., Kyle, 1985), which we define as *Splitting*. Second, we analyze traders' decisions to select the optimal trade size. For a given informed trader on a given day, we define *Intensity* as the ratio of the insiders' trade size, and the total volume for the same asset on the same date.

Our empirical results offer several novel perspectives on insider trading strategies. First, we show that almost 70% of insiders do not split their trades at all, which is at odds with predictions of a standard rational framework, in which some degree of splitting is always optimal. Second, we show that splitting is more likely when traders face greater information risk, which we approximate using unscheduled events (M&A). Splitting is also more prevalent for traders who suffer greater penalties ex post. Both effects are economically and statistically significant. Third, we find a more intensive trading when information risk is high though the result is not statistically significant. Further, trading intensity is positively and significantly related to ex-post monetary penalties.

In addition, we document a number of results associated with the traders' information environment and their trading profiles. First, we find that insiders use their private information faster when their private signal is more powerful and when they face more competition from other insiders. Further, strategic delays positively correlate with investors' trading experience. Second, the degree of splitting goes up when private signals are less powerful. In general, we find less splitting for large stocks, consistent with the notion of a lower price impact for large stocks. Third, insiders trade less intensely when either the tipper or the person conducting the trade works for the affected firm. Also, *Intensity* is negatively related to trading age, possibly indicating a positive correlation between age and risk aversion.

In our tests, we face a possible empirical challenge to identify legal risk because we use ex-post penalties as a proxy for the ex-ante risk. In particular, if penalties are associated with some traders' characteristics that also affect their trading strategies, our estimates could be biased. To alleviate this concern, we formally show that ex-post penalties do not relate significantly to a host of characteristics. We also show that time-invariant legal regime, measured by court fixed effects, cannot explain the effect of penalties on trading strategies. Nevertheless, omitted time-varying effects may still be at play.

To address this issue, we exploit a number of shocks to legal enforcement. First, we use the adoption of Dodd-Frank Act as a shock to the strength of legal enforcement. To motivate our experiment, we first document that in the wake of the new regulation, the observed Federal penalties became significantly more severe. We then show that, as a result of this change, insiders are more likely to split their trades and are less likely to submit large trades to the market. Both effects are economically and statistically significant.

We further exploit the cross-sectional variation in these effects by exploiting two important variations driving investment strategies: traders' sophistication and judicial power. In our first test, we exploit differences in traders' sophistication by comparing trades of insiders who are identified using the SEC Whistleblower Reward Program with trades of insiders identified through different means, e.g. exchange referrals, or brokers' tips. We argue that the former group of traders is likely more sophisticated since the regulator was not able to identify their activity. As such, one could expect their sensitivity to legal enforcement shocks may be different. This is what we find: Sophisticated insiders are more likely to split their trades in response to the shock; at the same time, their trading intensity is similar to that of unsophisticated investors.

In our second test, we exploit exogenous variation in individual courts' enforcement strength. Specifically, we use the case of Preet Bharara, the U.S. Attorney of the Southern District of New York (SDNY), as a shock to strength. Over the period of 2009-2013, Bharara has earned a reputation of a "crusader" prosecutor. Given that his employment spell at SDNY was ex ante uncertain we use the time period as a shock to perceived legal risk. Our identifying assumption is that traders located within the SDNY jurisdiction would perceive greater legal risk during this time. Consistent with the hypothesis, we find that traders from the SDNY region significantly reduced their trading intensity, both relative to the previous times and relative to traders from other jurisdictions. We find no evidence of changes in terms of trade splitting.

Another dimension of legal risk is the subjective probability of legal enforcement. We use Google trends associated with the phrase "illegal insider trading" as a shock to such a probability. Our results indicate that periods of high incidence of Google search are associated with a greater degree of splitting and lower degree of trading intensity. This result suggests that insiders do internalize the enhanced probability of legal enforcement when placing their trades.

Given the increase in the expected legal cost of insider trading, one would expect that agents focus on a subset of signals with higher expected return. We test this hypothesis by comparing the average signal strength for the sample of trades before and after each of the shocks we use. We find that indeed signal strength increases both in times of greater expected penalties and greater probability of legal enforcement. The effects are statistically and economically significant.

Our data also allow us to shed more light on the external validity of our findings. In particular, in our sample, we observe trades of investors who trade on private information but who are unlikely to perceive legal risk, which is confirmed by their court releases due to lack of support for illegal trading. We show that the sample of such traders behaves in a way that is consistent with them not acting under the threat of legal risk. They are less likely to split their trades and are more likely to trade more aggressively relative to a sample of traders facing legal risk. Overall, our evidence supports the view that insider traders internalize the increase in expected legal costs.

Our paper has important implications for price discovery. To shed some light on the process, we contrast the speed with which information is incorporated to prices in our data with the benchmark model of Kyle in which legal risk and information risk are absent. Our findings suggest that information gets incorporated faster in the framework with our two economic forces. This result indicates that information risk may be a stronger of the two forces when it comes to information revelation in prices.

Our paper relates to several strands of literature. First, we provide new direct evidence on what determines the use of private information in financial markets. Understanding the links between information flows and traders' behavior is important. Theoretical literature has identified links between private information and stock liquidity (e.g., Glosten and Milgrom, 1985; Kyle, 1985; Easley and Hara, 1987), option liquidity (e.g., Biais and Hillion, 1994), stock price volatility (e.g., Wang, 1993), and option price volatility (e.g., Back, 1993).

Second, we contribute to the literature on the workings of illegal insider trading. Early theoretical work by Ausubel (1990) studies the economic efficiency of insider trading regulation. Manove (1989) theoretically examines the ex-post consequences of insider trading for capital allocation in the economy. DeMarzo et al. (1998) and Carre et al. (2018)

derive conditions for the optimal design of insider trading regulation. On the empirical front, Meulbroek (1992) examines the impact of illegal trading on stock returns and market efficiency using a sample of legal cases from the 1980s. She shows that insider trades affect returns, as predicted by standard theory. Cornell and Sirri (1992) present a single-company case study of the impact of insider trading on stock liquidity. More recently, Del Guercio et al. (2017) study the effect of a time-varying enforcement environment on price discovery, Ahern (2017) studies insider traders' information networks, and Kallunki et al. (2018) analyze the impact of wealth on the decision to engage in insider trading.

More broadly, we contribute to a large literature that empirically analyzes financial misconduct. As examples, Dyck et al. (2010) analyze the behavior of whistleblowers in the context of corporate fraud. Karpoff and Lou (2010) discuss the importance of short sellers for the detection of financial reports' misrepresentations. Egan et al. (2018) analyze the sample of financial advisors and analyze ex-post penalties imposed on such advisors. Our paper is the first to focus on ex-ante implications of legal risk for trading behavior and information aggregation into prices.

Among the few studies that have examined flows of private information in financial markets are those of Koudijs (2015, 2016) using data from the 18th-century London and Amsterdam markets. While in Koudijs' framework one can plausibly identify the arrival time of private news, one cannot observe the precise nature of information or how individual traders use it in real time. These elements are crucial to our work. Also related is the work of Klein et al. (2017) who examine the characteristics of (legal) trades by corporate insiders.

The remainder of the paper proceeds as follows. Section 2 describes the sample of insider trading cases. Section 3 lays out the economic framework that generate testable hypotheses motivating the empirical part of the paper. Section 4 characterizes trading strategies and Section 5 discusses the determinants of insiders' trading strategies. Section 6 presents the empirical results related to information risk and legal risk. Section 7 provides a discussion of implications.

2 Insider Trading Background and Data

2.1 Background on Insider Trading

Insider trading is a term that includes both legal and illegal conduct. The legal variety is when corporate insiders—officers, directors, large shareholders, and employees—buy and sell shares in their own companies and report their trades to the SEC. Legal trading also includes, for example, someone trading on information he or she overheard between strangers sitting on a train or when the information was obtained through a non-confidential business relationship. On the other hand, illegal insider trading (IIT) refers to "buying or selling a security in breach of a fiduciary duty or other relationship of trust and confidence, while in possession of material, nonpublic information about the security."

The legal framework prohibiting insider trading was established by Rule 10b-5 of the Securities Exchange Act of 1934. Under the classical view of insider trading, a trader violates Rule 10b-5 if he trades on material, nonpublic information about a firm to which he owes a fiduciary duty, where information is deemed "material" if a reasonable investor would consider it important in deciding whether to buy or sell securities. The misappropriation theory applies to prevent trading by a person who misappropriates information from a party to whom he or she owes a fiduciary duty—such as the duty owed by employee to employer or by lawyer to client.

The consequences of being found liable for insider trading can be severe. Individuals convicted of criminal insider trading can face up to 20 years imprisonment per violation, criminal forfeiture and fines up to $5,000,000 or twice the gain from the offense. A successful civil action by the SEC may lead to disgorgement of profits and a penalty not to exceed the greater of $1,000,000, or three times the amount of the profit gained or loss avoided. In addition, individuals can be barred from serving as an officer or director of a public company, or in the case of licensed professionals such as attorneys and accountants, from serving in their professional capacity before the Commission. It is also not uncommon for individuals or companies involved with government insider trading actions to face private suits.

2.2 A Tale of an Insider Trader

To illustrate the main elements of a typical insider trading case, we begin with the story of Matthew Martoma–one of the most prominently featured insiders over the last few decades. Martoma worked at CR Intrinsic, an unregistered investment adviser, between 2006 and 2010, serving as a portfolio manager from at least January 1, 2008 until his departure in 2010. He perpetrated the insider trading scheme with Sidney Gilman, a professor of neurology at the University of Michigan. Gilman served as a consultant to Elan (ELN) and Wyeth (WYE)–two pharmaceutical companies from 2003 until 2009.

Between 2006 and 2008, Elan and Wyeth jointly conducted a Phase II clinical trial for a potential drug to treat Alzheimer's disease called bapineuzumab. Gilman, who served as a consultant for Elan, had continuing access to material nonpublic information concerning the Phase II Trial. Among many duties, Gilman agreed to present, on behalf of Elan and Wyeth, the Phase II Trial results at the International Conference on Alzheimer's Disease (the "ICAD"), a medical conference that was scheduled to be held on July 29, 2008. As a result of agreeing to serve as the presenter, Gilman was given access to the full Phase II trial results approximately two weeks prior to the July 29 announcement. By virtue of his roles in the clinical trial, Gilman owed Elan a duty to hold in strict confidence all information he learned in connection with his participation in the clinical trial and to use such information only for Elan's benefit.

Gilman first met Martoma through paid consultations arranged by the expert network firm. Gilman provided Martoma with material nonpublic information concerning the Phase II Trial starting in at least 2007. In the weeks leading up to the July announcement, Gilman held several calls with Martoma during which he provided Martoma information regarding the safety and efficacy results for the Phase II Trial. On July 17-18, 2008, Gilman and Martoma had a lengthy phone call during which Gilman provided Martoma with confidential information regarding the detailed results of the Phase II Trial, which suggested that the drug had potential serious side effects.

On the morning of Sunday, July 20, 2008, following his calls with Gilman, Martoma indicated to a portfolio manager of the affiliated asset management company that he was no longer comfortable with the Elan investments held by the CR Intrinsic and the manager's portfolios. Before the market opened on July 21, 2008, these portfolios held over 10.5 million Elan securities worth over $365 million and over 7.1 million Wyeth shares worth over $335 million, for a total position size of over $700 million. On Monday, July 21, 2008, both companies began selling Elan and Wyeth securities they held. Later, the manager communicated to Martoma that they executed a sale of over 10.5 million ELN at an average price of $34.21. This trade was executed quietly and efficiently over a 4-day period through algos and darkpools. Martoma urged the manager to sell the Elan securities in both portfolios quickly.

In total, between July 21, 2008 and July 29, 2008 (the last trading day before the post-market July 29 announcement), both companies sold over 15 million Elan securities for gross proceeds of over $500 million. Although the investment advisers' portfolios achieved a zero balance in Elan securities by July 25, 2008, they continued to sell short Elan securities until the announcement. By the close of the market on July 29, 2008, both companies had a combined short position of approximately 4.5 million Elan securities. The trading in Elan securities constituted over 20% of the reported trading volume in the seven days prior to the announcement. In addition, between July 21, 2008 and July 29, 2008, both companies sold over 10.4 million shares of Wyeth for gross proceeds of over $460 million, including over 6.1 million Wyeth shares worth over $270 million during the very day of the announcement. As a result of these sales, they had a zero balance in Wyeth stock during the trading day on July 29, 2008, but continued to place short sales that day. By the close of the market on July 29, 2008, the portfolios had a combined short position of approximately 3.3 million Wyeth shares. The trading in Wyeth securities constituted over 11% of the reported trading volume in the seven days prior to the announcement. In addition, on July 28 and July 29, they purchased over $1 million worth of Elan put options with strike prices below the Elan share price on those trading 3 days. Figure 3 presents detailed chronology of the major positions and trades related to the case.

On July 29, 2008, after the close of the U.S. markets, Gilman presented the results of the Phase II trial at the ICAD. Although Elan and Wyeth emphasized the positive aspects of the trial the market reacted negatively to the full results. On July 30, 2008, the first trading day after the announcement, Elan's share price fell from $33.75 (the closing price on the day of the announcement) to $19.63 (the closing price on the day after the announcement), a decline of nearly 42%. Similarly, Wyeth's stock price fell from $45.11 to $39.74, a decrease of nearly 12%. Figure 4 presents the time-series evolution of the stock prices for both companies.

As a result of the trades that were entered into during the period between Martoma's conversation with Gilman on July 17, 2008 and the public announcement, both companies reaped profits and avoided losses of over $276 million. At the end of 2008, Martoma received a bonus of over $9.38 million. Gilman received over $100,000 from the expert network firm for his consultations with Martoma and other related analysts. In contrast to 2008, Martoma was unable to generate such outsized returns in 2009 and 2010, and did not receive a bonus in either of those years. In a 2010 email suggesting that Martoma's employment be terminated, an officer from his company stated that Martoma had been a "one trick pony with Elan." Following the inquiry based on the whistleblower tip, the SEC has launched a formal investigation. On February 6, 2014, Martoma was found guilty of his insider trading. Subsequently, on September 8, 2018, the Southern District of New York sentenced him to 9 years in prison. He was also requested to forfeit his bonus. Gilman agreed to forfeit $234,000–the amount he earned in the scheme, plus interest.

2.3 Data Collection

We retrieve a list of SEC investigations from all SEC press releases that contain the text "insider trading." We use this list to obtain all the available civil complaint files available on the SEC website. In cases in which the complaint file is not available on the SEC website, we rely on manual web searches and on information from the U.S. District Court where the case is filed. We collect all files starting from January 2001 until December 2015. We track all documents that provide updates on a previously released legal case. Whenever updated information is made available at a later date, we rely on the most recent version.

The resulting sample represents all SEC cases that were either litigated or settled out of court. Most complaint files include a detailed account of the allegations. Since the documents provide most of the relevant information in textual form, the data files must be thoroughly read and summarized by hand. Available information typically includes biographical records of defendants, individual trades, a description of the leak that the trades are linked to, as well as the relationships between tippers and tippees.

We organize the information by characterizing trades and information events. A *trade* is any single transaction record for which we can observe a date and a trading instrument (e.g., stocks or options). For most trades, information about the price, trade direction, quantity, trading profits, and the closing date of the position are also available; as well as the contract characteristics for options. Whenever only a date range is available, we only consider as trading dates the first and the last day of the range. This condition reduces the potential number of trading dates but yields well-identified trading date records throughout the analysis. We also record individual names in cases in which more than one person/firm executed trades on a single piece of news.

An *information event* is a collection of one or more trades that were motivated by a unique piece of private information, such as an earnings announcement or a merger. For our purpose, the key information event records include companies involved, the nature of the leaked information (e.g., a new product), and the date at which the information was released to the general public. We also collect information on the date of information transmission from tipper to tippee. This information allows us to test hypotheses on strategic trading delays.

2.4 Descriptive Statistics

Our data cover 453 legal cases. The event types are described in Table I. The most frequent events are M&As (55.90%), followed by earnings announcements (15.06%). The business events and corporate events categories (10.71%) include, among others, items such as information about products, a firm's projects, patents, FDA medical trials, corporate restructuring, bankruptcy, and fraud. The average number of cases per year in our sample is 30.83, with a maximum of cases (46) filed in 2012. The distribution of the number of firms per case is highly asymmetric. Approximately 80% of the cases involve a single firm, while 4% of the cases involve 10 firms or more. We identify a total of 5,058 unique trades involving 615 firms. The vast majority of trades are executed via stocks (67.06%) and options (31.83%). The remaining few are trades in American depositary shares and bonds. There are 4,220 buys (83.43%) and 838 sells.

Even though our legal cases date back to 2001, several cases involve trades that took place earlier on. Consequently, our sample of trades spans a longer time period, 1995–2015. The sample is quite evenly distributed over time, with over 100 trades in each year between 1999 and 2014. We observe a smaller number of trades in the 1990s and again toward the end of our sample, in 2015. The latter situation is explained by the delay with which cases can be identified and formally prepared by the SEC. The observed dispersion of trades across years is an attractive feature of our data that allows us to address common identification issues, such as time-specific macro events, etc. The distribution of trades is highly dispersed across many different industries. The three most represented industry sectors in our sample are chemicals, business services, and electronic equipment, which account for more than 40% of all trades. However, we note that the trading involves companies coming from almost all industrial sectors.

In Table II, we summarize our data at the trade level, which is our main unit of observation. The median time between the arrival and the use of information by insiders is two days. In turn, the median number of days from a trade till an information event is seven days. The majority of cases involve single trades in a given company, but a subset of traders execute more than one trade. The median horizon between the first and the last such trade is eight days. Further, a median trader in our sample executes 10 trades, with a maximum of 97 trades. A median firm receives 13 trades and a median legal case involves two firms. The median age of tippers and traders is almost identical and equals 45 years and 46 years, respectively. The vast majority of more than 90% of tippers and traders in our sample are male. The profits reported by traders are highly skewed, with an average trade profit of $1.01 million and the median of $90,000, and 49% of trades elicit at least $100,000 in profits.

3 A Simple Framework

In this section, we analyze a simple theoretical framework to motivate our predictions about the behavior of informed traders and the information aggregation into asset prices.

3.1 Information and Enforcement Environment

We consider a two-period strategic trading model in the tradition of Kyle (1985). The value of the asset v satisfies $v \sim N(p_0, \sigma_v^2)$. One informed trader observes v at time 0 and submits market orders x_t, $t = 1, 2$. Nonstrategic liquidity traders submit market orders of size u_t with $u_t \sim N(0, \sigma_{u_t}^2)$. A competitive market maker observes the aggregate order flow as given by $y_t = x_t + u_t$ and sets the asset price p_t accordingly.

Besides an adverse realization of uninformed orders u, the informed trader faces two additional sources of risk. First, there is *information risk*. With probability ρ, the value v is publicly announced between periods 1 and 2 and the informed trader loses his informational advantage after one period.

Second, the informed trader internalizes *legal risk*. With probability Q the regulatory agency detects the illegal trade and, in that case, assigns a monetary penalty C. The expected penalty, K, is thus $Q \times C$. We assume that enforcement takes place after trades, that is, any legal penalty is known after period 2. We make the following additional assumptions about the enforcement primitives.

Assumption A1. *The ex-post probability of detecting insider trade* x_t, $t = 1, 2$, *equals* $Q(x_1, x_2) = [q(x_1) + q(x_2) - q(x_1)\, q(x_2)]$, *where*

$$q(x_t) = \frac{|x_t|}{a^2} 1_{x_t \in [-a,a]} + 1_{x_t \notin [-a,a]}, a > 0.$$

Assumption A2. $C(x_1, x_2) = c(|x_1| + |x_2|)$, $c > 0$.

We briefly comment on these assumptions. A1 implies that the probability of detection depends on the trade size in each period. Note that $q(0) = 0$ and that $q'(x) > 0$ is constant for $x < |a|$. One can interpret this connection as originating in a broker reporting to a regulatory body such as the SEC or FINRA if a customer order is suspicious. The larger the absolute value of the order, the more likely the order is deemed suspicious. The informed trader is always caught provided the order size is too large, that is, $x > |a|$. From the perspective of the regulator, and consistent with trade anonymity, the probability of detection in period t does not depend on trades in period t'. However, if any insider trade is detected, analysis of the insider's trade account reveals the sequence $\{x_1, x_2\}$. A2 captures the intuitive fact that the regulator imposes penalties that are proportional to the economic importance of the illegal trade.

Furthermore, we note that information and legal risk are likely to imply different trade paths. If information is lost after period 1, one would not anticipate $x_2 > 0$. However, if the information is not lost, regardless of the ex-post investigation outcome, the insider is likely to trade in each period. In other words, the regulator can impose ex-post penalties on insider trading but cannot prevent trades in either period.

3.2 Trade Dynamics and Equilibrium

The insider trader value functions are as follows. At the beginning of period 1, having observed v, the informed trader has a value function V_1 given by

$$V_1 = \max_{x_1 \in \mathbb{R}} \left\{ \mathbb{E}\left[(v - p_1)\, x_1 + (1 - \rho)\, V_2 | v \right] - \rho K\left(x_1\right) \right\}. \tag{1}$$

The third term in equation (1) corresponds to the expected penalty in the event that information is disclosed early and trading in period 2 becomes worthless, where $K\left(x_1\right) = q(x_1)C(x_1)$. Conditional on no early disclosure, the value function in period 2 is given by

$$V_2 = \max_{x_2 \in \mathbb{R}} \left\{ \mathbb{E}\left[(v - p_2)\, x_2 | v, p_1 \right] - K\left(x_2 | x_1\right) \right\}. \tag{2}$$

Competition and informational efficiency force the market maker to set prices $p_1 = p_0 + \mathbb{E}\left[v|y_1\right]$ and $p_2 = p_1 + \mathbb{E}\left[v|y_1, y_2\right]$.

In what follows, we assume that a is large enough so that the informed trader find it optimal to select $x_t < |a|$ and so that the product of detection probabilities is of second order, $q(x_1)\, q(x_2) \approx 0$. The following proposition characterizes the resulting linear equilibrium.

Proposition 1. *There here is a unique linear equilibrium with $x_t = \alpha_t + \beta_t v$ and $p_t = p_{t-1} + \lambda_t (x_t + u_t)$, $t = 1, 2$, given by*

$$\beta_1 = \frac{\left(1 - \left(\frac{1-\rho}{2\gamma_2}\right)\left(\lambda_1 + \frac{2c}{a^2}\right)\right)}{2\left(\gamma_1 - \left(\lambda_1 + \frac{2c}{a^2}\right)^2\left(\frac{1-\rho}{4\gamma_2}\right)\right)}, \quad \alpha_1 = -p_0\beta_1, \tag{3}$$

$$\beta_2 = \frac{1}{2\gamma_2}\left(1 - \frac{2c}{a^2}\beta_1\right), \quad \alpha_2 = \frac{1}{2\gamma_2}\left(\frac{2c}{a^2}\beta_1 p_0 - p_1\right), \tag{4}$$

$$\lambda_1 = \frac{\beta_1\sigma_v^2}{\beta_1^2\sigma_v^2 + \sigma_u^2}, \tag{5}$$

$$\lambda_2 = \frac{\beta_2\sigma_v^2}{\left(\beta_2^2 + \beta_1^2\right)\sigma_v^2 + \sigma_u^2}, \tag{6}$$

$$\gamma_t = \left(\lambda_t + \frac{c}{a^2}\right). \tag{7}$$

Proposition 1 allows us to investigate the connections between the information and legal environment and the behavior of insiders and market makers, $\{\beta_t, \lambda_t : t = 1, 2\}$. These relations are illustrated in Figure 5. Next, we exploit such connections to derive empirical predictions on prices and trade quantities.

3.3 Empirical Predictions

To illustrate the implications of the model's equilibrium, we study the average trading behavior of the informed trader using the following metrics:

$$\text{Intensity} = \frac{1}{2}\mathbb{E}\left(|x_1| + |x_2|\right), \tag{8}$$

$$\text{Splitting} = \mathbb{E}\left(1 - \frac{|x_1 - x_2|}{|x_1| + |x_2|}\right), \tag{9}$$

$$\text{Duration} = \mathbb{E}\left(\frac{|x_1| + 2\,|x_2|}{|x_1| + |x_2|}\right). \tag{10}$$

Intensity captures the average informed trade volume. *Splitting* reflects how even informed trades are over time and takes a value equal to 0 if all trades are concentrated in a single period and a value equal to 1 if trades are identical. *Duration* reflects the proportion of early/late trade volume, taking a value between 1 and 2.

We simulate trading sessions that allow us to compute the moments defined in (8)–(10) assigning parameter values as follows. The median information horizon in the sample equals two weeks and, therefore, we interpret a period as representing a trading week. We set the standard deviation of noise trading equal to the standard deviation of stock volume at weekly frequencies, $\sigma_u = 67.49$. To capture the notion of 'suspicious' order size, we normalize the enforcement detection parameter a relative to σ_u, $a = \sigma_u$. Based on A2, informed orders larger than $|\sigma_u|$ are automatically flagged. To calibrate σ_v, we consider a representative stock price of $p_0 = 10$ and find the standard deviation value that rationalizes an absolute value return of $\overline{R} = 30.61\%$, as observed in the sample. We do so by inverting the mean equation of the folded normal distribution,[2] obtaining $\sigma_v = 8.43$. For expositional clarity, in the simulations that follow, we condition the moments on paths for which there is no early information disclosure.

[2]Recall that if $x \sim N(\mu, \sigma)$, $y = |x|$ has a mean value $\mu_Y = \sigma\sqrt{\frac{2}{\pi}}e^{\left(-\mu^2/2\sigma^2\right)} + \mu\left(1 - 2\Phi\left(\frac{-\mu}{\sigma}\right)\right)$.

Legal Risk To develop economic intuition, consider first the case where the insider faces legal but no information risk ($c > 0$, $\rho = 0$). The left panel of Figure 1 displays the average informed trade size[3] and asset price by period and the right panel displays the strategic metrics (8)–(10). As the penalty parameter c increases, the insider internalizes expected enforcement costs by reducing the order size in each period and, in equilibrium, the price is less informative about the private signal. Because trade sizes become smaller, the relation between Intensity and c is negative. Note, however, that the effect on trade sizes is not identical. Indeed, the penalty as a factor that increases the effective impact of an informed trade, thereby incentivizing more even trades over time, thus increasing the value of *Splitting*. As the relative weight of trades in $t = 1$ increases, the value of *Duration* decreases.

Information Risk Consider now the case where the insider faces information risk but no legal penalty ($\rho > 0$, $c = 0$). The left panel of Figure 2 displays the average informed trade size and asset price by period and the right panel displays the strategic metrics (8)–(10). As the risk of losing the information advantage increases, the insider trades more aggressively in the early trading round and less aggressively in the late one. In the aggregate, the informed trade volume increases and *Intensity* displays higher values. These trading patters imply that the value of *Splitting* increases for relatively low values of ρ and decreases for relatively high values. The effect on Duration, on the other hand, is monotonic: the greater the risk the greater the importance of informed trading in the first period.

Economic Environment Besides legal and information risk, the economic environment is defined by three primitives that are also present in the traditional Kyle model: the value of the private signal, $(v - p_0)$, the volatility of noise trading, σ_u, and the ex-

[3]We note that the values of x_1 and x_2 on the vertical axis, with zero legal penalty, corresponds to the canonical two-period Kyle equilibrium.

Figure 1. Empirical Predictions: Legal Risk

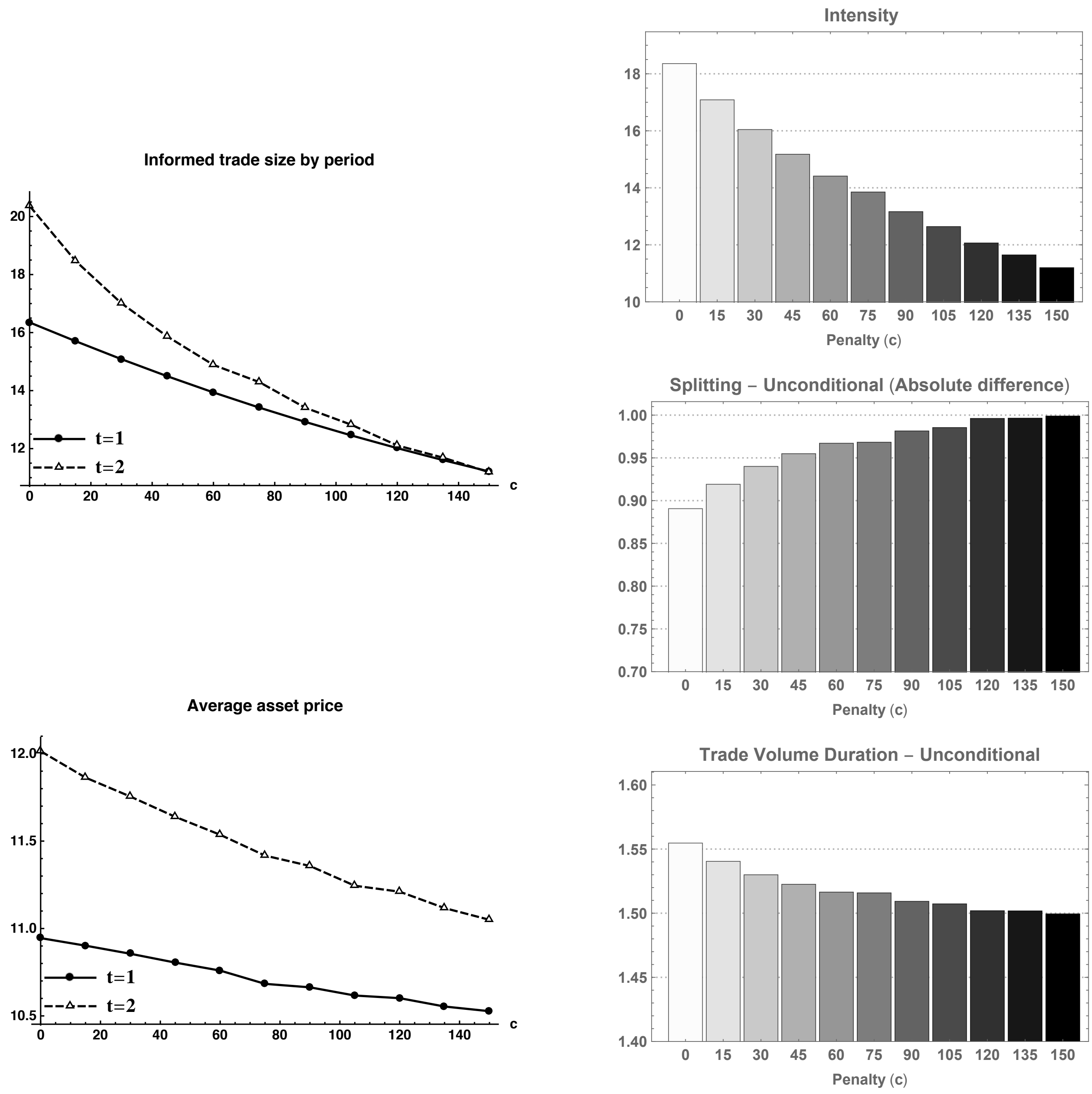
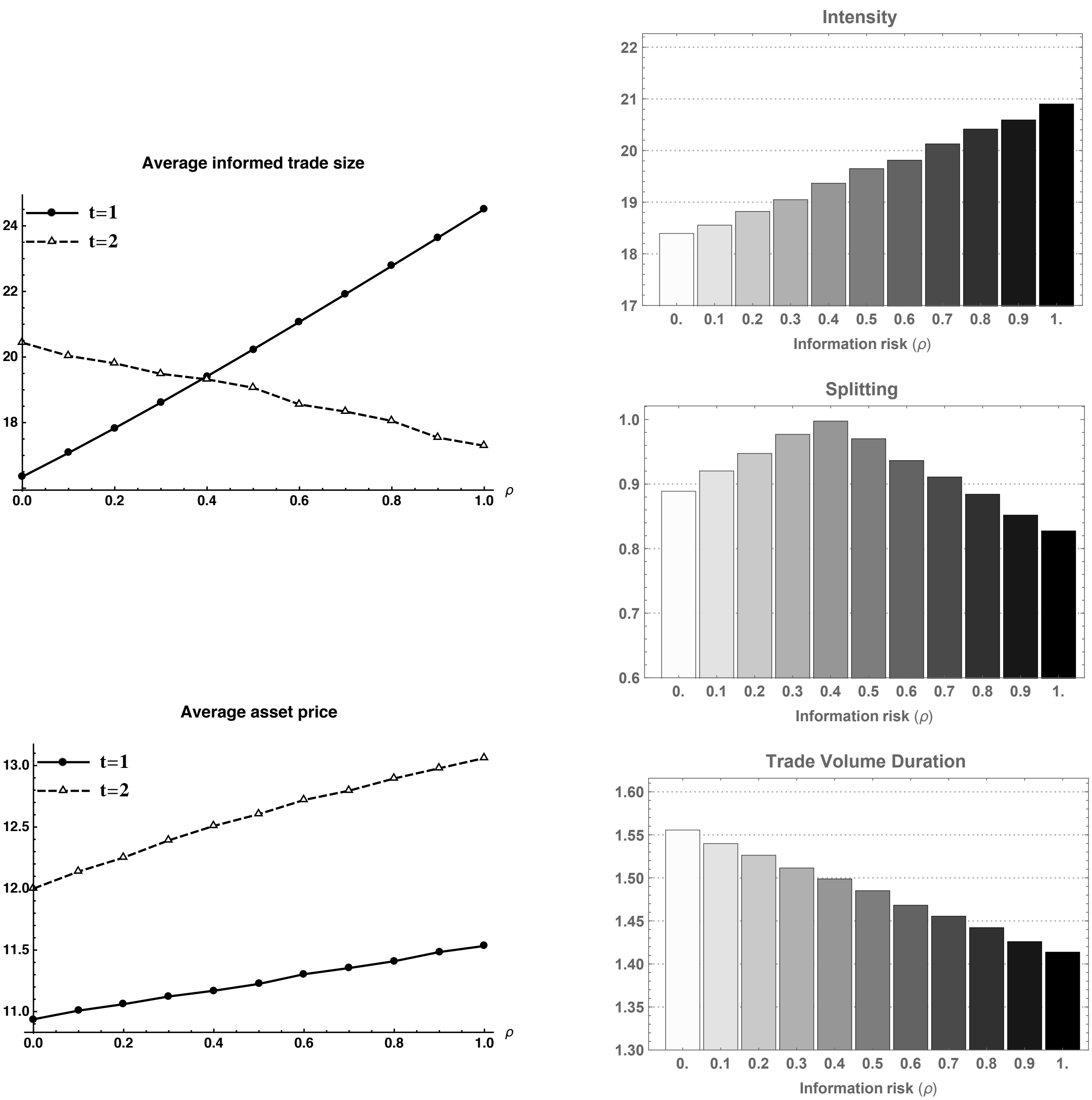


Figure 2. Empirical Predictions: Information Risk



ante volatility of the asset value, σ_v. We briefly comment about the impact of parameter changes on the summary statistics (8)–(10). As one would expect, an increase in the value of the private signal increases the average trade size in each period and, therefore, increases *Intensity*. The effect on *Splitting* is less significant though, since the relation between β_1 and β_2 is not driven by the realization of v. As expected, and increase in σ_u and σ_v have a positive and a negative effect on *Intensity*, respectively. The effect on *Splitting* is also of opposite sign: an increases in σ_u decreases the value of *Splitting* while an increase in σ_v induces the informed trader to split trades more evenly.

4 Insider Trading Strategies

In this section, we define the key aspects of trading strategies potentially used by insiders.

4.1 Methodology

The case of Martoma indicates a number of key elements that are generally associated with any insider trading case. In Figure 6, we provide the graphical presentation of the timing of a trading strategy. We note a number of important dates related to information transmission and trading events.

We first consider information transmission. T_{info} denotes the date at which a trader receives the signal about firm fundamentals. Directly linked to this date is T_{public}, the date at which such information becomes publicly available. Next, we focus on trading events. T_{first} and T_{last} are the first and last dates at which an insider in a given company trades on private information before T_{public}. T_{close} denotes the date at which the trader closes the initial position to realize profits.

We take both T_{info} and T_{public} as exogenous parameters from the trader's perspective as this allows us to benchmark individual investors' trading decisions. Given $\{T_{\text{info}}, T_{\text{public}}\}$, the trader decides $\{T_{\text{first}}, T_{\text{last}}, T_{\text{close}}\}$ endogenously. Accordingly, we define the exogenous information horizon as $H_{\text{info}} = T_{\text{public}} - T_{\text{info}}$ and the endogenous trading horizon as $H_{\text{trade}} = T_{\text{last}} - T_{\text{first}}$, which is measured in number of trading dates.

As a next step, we define a number of trading strategies that constitute the bulk of our empirical tests and are consistent with the measures defined in our theoretical framework. Our first measure relates to trade splitting. In executing their ideas, traders must decide whether to split their trades. In principle, strategic investors distribute their trades over multiple periods (e.g, Kyle, 1985), especially if the trading costs are high. On the other hand, the presence of information risk may deter traders from distributing their trades more equally over multiple trading days. With this goal, we define *Splitting* as:

$$\text{Splitting:} = \frac{T_{\text{last}} - T_{\text{first}}}{H_{\text{info}}}.$$

Apart from timing his trades, a trader needs to select a trade size. For a given informed trader j, in company i , on day t, we define the ratio between the informed trade size, and the total volume on that date, *Ratio*:

$$\text{Ratio}_{i,j,t} := \frac{\text{informed trade volume}_{i,j,t}}{\sum_j \text{Volume}_{i,t}}.$$

By construction, the value of *Ratio* varies between 0 and 1 (when trader j is a counterparty in every trade). The average ratio is simply given by $\text{AvgRatio}_j = \frac{1}{\tau} \sum_{i=1:I} \text{Ratio}_{i,j,t}$, where τ represents the number of informed trading dates. Our final measure is maximum intensity aggregated across different trading instruments and defined as Intensity $= \max_{\text{a}} \left\{ \frac{\text{Informed volume}_a}{\text{Total volume}_a} \right\}$, where $a \in \{\text{stocks, calls, puts}\}$.

4.2 Empirical Characterization of Trading Strategies

We next characterize the distribution of the trading strategies in the sample of our insider traders. We first focus on the extensive margin of *Splitting*. In Table III, we show that the incidence of zero values is economically significant. The proportion of cases in which investors trade only in one day equals 58.2% for earnings announcement cases and 74.5% of M&A cases. This fact is robust across both stocks and options markets. Further, the proportion of zero values for *Splitting* is similar for both sophisticated and unsophisticated traders.

We further analyze the intensive margin of our trading measures. Specifically, Figure 7 displays the histograms and kernel densities (in red) of *Splitting* (Panels a and b) and *Intensity* (Panel c). We report both raw and normalized versions of *Splitting*. For *Intensity*, we show the maximum across all asset types.

We observe that the distributions show rich cross-sectional variation in trading behavior, both Splitting and Intensity. The variation is particularly visible in the case of normalized measures. In unreported results, we observe that trading intensity is more dispersed for options compared to stocks. We explore what determine these decisions in the next section.

5 Determinants of Insider Trading

Following the seminal contribution of Becker (1968), we model the trading decision by insiders as a rational response to the tradeoff between trading benefits and costs related to the trading environment, and the individual traders' characteristics. In this section, we discuss most relevant dimensions along which investors evaluate their trading decisions. The major two we highlight are information risk and legal risk. In addition, we relate the trading decision to various aspects of information environment–signal strength, reliability, amount of noise trading, and competition among insiders– and trader characteristics–gender and age.

5.1 Information Environment

Signal Strength Our empirical design relies on the work by the SEC to verify the material and non-public nature of information. Naturally, an interesting issue arises: How 'material' is the information received? In other words, how strong is its information content? To shed light on this aspect, for each information event, we compute the percentage change in the corresponding stock price from the open of the day of the informed trade to the open price immediately after the information becomes public. For example, if information about earnings is disclosed overnight at date t, we consider the open price at date $t+1$. Table IV shows the results for each type of news and the aggregate sample.

Information power or strength may increase the opportunity cost of not trading before the public release. Consequently, we posit that information that is perceived as stronger, ceteris paribus, should (i) decrease *Splitting* and (ii) increase *Intensity*.

We assume that the trader receives from the tipper a signal containing both the direction of the news and an indication of relevance. Even when only the direction and the type of information is provided, a trader may form accurate beliefs about how much information is on the tip. Such beliefs, however, are not directly observable. Accordingly, we define an ex-post measure, *Strength*, as follows:

$$\text{Strength:=}\frac{\text{Opening Price}(T_{\text{public}}+1)-\text{Opening Price}(T_{\text{first}})}{\text{Opening Price}(T_{\text{first}})}.$$

Given the (at least partial) endogeneity of the timing of the first trade, we also consider

$$\text{Alt Strength:=}\frac{\text{Opening Price}(T_{\text{public}}+1)-\text{Opening Price}(T_{\text{info}})}{\text{Opening Price}(T_{\text{info}})}.$$

For positive news, the average return is greater than 43.5% and the median return is approximately 33.7%. These values are remarkably large, given that the median time interval from a trade to private information disclosure is a mere seven days. Moreover, one could treat these numbers as a lower bound on the true signal strength, since about 30% of trades are in options, hence embedding leverage. To put these numbers in another perspective, we construct benchmark returns for a sample of SEC 13D form filers between 1994 and 2014. The benchmark return is based on the return measured from the opening of the day when the 13D filer trades an asset until the opening of the day following the release of the trade information to the public. The trades of 13D filers represent significant long positions in a security and have been shown to predict positive stock returns, so they can be interpreted as based on positive news (e.g., Brav, Jiang, and Kim, 2015; Collin-Dufresne and Fos (2015)). The mean and median returns for 13D filers are 4.9% and 2.4%, respectively.

Information Reliability Another factor we consider is the precision of information (information reliability). Intuitively, information precision reduces the incentive to wait for subsequent signals to arrive and thus increases the opportunity cost of not trading before T_{public}. Furthermore, if traders are risk averse, more precise signals would induce larger trade sizes. This leads us to the following hypothesis. Information that is perceived as more reliable, ceteris paribus, should (i) decrease *Splitting* and (ii) increase *Intensity*. We deem information as more reliable when its source comes directly from the affected firm ("first hand" information). Accordingly, we define *Tipper Insider* as an indicator variable that equals one if the trader or the tipper works for the affected company; and zero, otherwise.

Fundamental Uncertainty One of the key factors affecting trading is the amount of fundamental uncertainty. In the presence of high uncertainty, investors may have an incentive to resolve uncertainty earlier, that is, not split their trades. They may also respond more to their underlying signals by trading larger quantities. In the paper, we measure fundamental uncertainty, *Realized Volatility*, using volatility of stock returns, measured over a short 30-minute window.

Noise Trading The decision to trade by insiders is also a function of the amount of noise trading in the market. An increase in noise trading should make it easier for insiders' to place their trades, thus they should be trading relatively larger quantities of assets. It should also make them smooth their trades more due to more favorable investment opportunities. We measure the amount of noise trading using the volatility of daily trading volume in a given month, *Volume Vol*.

Liquidity Costs The main driving force underlying Kyle, 1985 is market liquidity. Traders may decide to split their trades or trade less aggressively if their trades reveal more information to the public, that is, they have price impact. In this paper, we aim to measure the variation in such price impact using the market capitalization of the company insiders trade. Specifically, we use the natural logarithm of the market cap, *Ln (mkt. cap)*. Notably, we consider this transaction force as a benchmark case and generate deviations therefrom by looking at information risk and legal risk.

Trader Characteristics Apart from variables defining information environment, we also consider individual characteristics of traders in our sample. Prior research shows that trading behavior depends on individual traders' characteristics, such as age, gender, wealth, or occupation. The same characteristics could be also related to insider trading behavior. If age is a proxy for individual risk aversion then one could expect that older traders should behave more conservatively, holding everything else constant. At the same time, if age is a proxy for career concerns, then older traders may in fact exhibit greater risk tolerance to trading than younger traders.

Likewise, various sociological studies argue that male are on average more overconfident than female. As such, they should be less sensitive to penalties and any other costs than female. The only caveat of looking at the data in our paper is that our sample is predominantly male dominated so we may not have enough statistical power to identify any effects due to gender. Finally, we argue that the expertise of insiders may affect their propensity to hide trades. As such, we should expect such traders to split their trades more and trade lower quantities of assets. In Table V, we show the distribution of professions across insiders in our sample.

Our empirical measures are, *Age*, traders' age in years, and traders' gender. *Gender* is an indicator variable equal to one if a trader is male, and zero, if a trader is female. *Experience* is equal to one when the trader works in the financial services industry or is a top executive in any industry (defined as vice president or higher), and zero, otherwise. We summarize all the characteristics pertaining to information environment in Table VI.

5.2 Information Risk

One of the key drivers of trading strategies we explore in this paper is information risk, that is, the risk that information may become public at a future stochastic time. To the extent that the revelation time is unpredictable one would expect that insiders would act on their information faster relative to the case with no information risk, holding everything else constant. In our empirical tests, we use trades associated with M&A activity as more likely subjected to such risk. To this end, we define an indicator variable *Information Risk* equal to one for trades associated with M&A activity, and zero for all other trades.

As an alternative way to measure information risk we look at the number of traders, *Traders*, involved in a given SEC case. Informed trading behavior is likely affected by the intensity of competition in the use of information. Indeed, economic intuition suggests that the speed of information use should increase when there is a threat of losing it to a competitor (e.g., Holden and Subrahmanyam, 1992). Hence, we argue that traders acting on information also disseminated to other investors may be less likely to delay their trades.

To assess whether the two variables broadly reflect the reduced incentive to hold off private information, we relate them to a measure of *Strategic Delay*, that measures the span from the time an insider receives information till the time he/she uses it. Formally, we define the strategic delay as $T_{\text{first}} - T_{\text{info}}$, as well as its normalized version, Strategic Delay $\in [0, 1]$, given by

$$\text{Strategic Delay} := \frac{T_{\text{first}} - T_{\text{info}}}{H_{\text{info}}}.$$

If M&A activity and/or competitive trading environment indeed capture information risk, one would expect to see a negative relationship between *Information Risk (Traders)* and *Strategic Delay.* Formally, we estimate the following regression model

$$\text{Strategic Delay}_i = a{+}b{\times}\text{Controls}_i\ {+}\text{c}{\times}\text{Information Risk} + \text{d}{\times}\text{Traders} + \text{error}_i.$$

where *Controls* is a vector of controls including Strength, Realized Volatility, Volume Volatility, Tipper Insider, Ln(mkt. cap), Age, and Gender. Our coefficients of interest are c and d. We present the results in Table VII. In column (1), we estimate the regression model without any controls. Both coefficients of c and d are negative, but only the estimate of c is statistically significant. In columns (2) and (3), we extend the empirical specification by adding two sets of controls. Again, we obtain very similar results. The effect of *Information Risk* is negative and statistically significant and the effect of *Traders* is negative, but insignificant. We also find economically relevant results for the set of controls we use. The results indicate that traders use their information more quickly when their private signals are more powerful. On the other hand, traders delay the use of information for stocks with high realized variance. These results are consistent with our earlier hypotheses. At the same time, *Tipper Insider* and *Volume vol* are not strong predictors of the information timing decision. In the subsequent tests, we only use *Information Risk* as a primary variable of interest.

5.3 Legal Risk

One of the important considerations for illegal insider traders is legal risk. This risk can materialize through two types of penalties: pecuniary ones, which are typically assigned in the process of civil court investigation, and non-pecuniary ones, which can be additionally applied through the criminal investigation. Typically, monetary penalties are a function of the profits realized in the investigation. The common rule is that the penalty be equal to the size of the profits realized in the case, in addition to the forfeiture of profits realized in the case. In addition, the court requires the insiders to pay interest on the amount of profits they make and typically bars a convict from trading activity for a substantial time period. Criminal cases usually end up with two types of resolutions: jail penalty and probation. Jail penalty varies from as little as a few months to as several years. Probation usually does not take longer than 3 years. Occasionally, a case presented by the SEC can be dismissed by court due to lack of evidence that would comply with the rule of insider trading. Notably, these are usually politically popular cases, or cases with a significant amount of insider trading (like the SAC fund).

We collect detailed information on both types of penalties for each individual trader in our sample associated with a given investigation. Some of the information is reported in the complaint files reported by the SEC. However, a significant fraction of files misses the information. We append such information through through website searches, including court reports, newspaper articles, and legal websites (Lexis Nexis). We summarize the data in Table VIII below.

The average monetary penalty for a given trader in our sample amounts to $2.85 million, while the median is $0.2 million. The largest monetary penalty has been assigned to Raj Rajaratnam of Galleon Fund and equals approximately $160 million. Many cases involve more than one trader. As such, a given case can bring a significantly larger amount of money. The total penalty per case in our sample equals $11.74 million, with the median of $1.25 million. Further, penalties can vary across traders in a given case. The average standard deviation of within case penalties equals $3.16 million, though the median is significantly smaller and equal to $0.08 million. More than 10% of traders in our sample receive a jail penalty, with an average duration of 3.5 years. In a given case, 10% of traders are jailed. An additional 23.5% of traders receive probations. Finally, 12% of traders have their cases dropped.

The severity of penalties may depend on the individual courts that rule a particular case. The assignment of cases to courts is generally based on the geographic proximity to traders' permanent address. We collect details of the different courts involved in the cases. The sample includes a total of 77 different courts. The most prominently featured courts include the Southern District of New York (30.4% of cases), District of Columbia (6.1%), and Central District of California (5.5%).

6 Cross-Sectional Analysis

In this section, we present the results on the role of various predictors of strategic trading decisions. First, we show the results from our basic regression to illustrate the interplay between information risk and legal risk. Next, we introduce various shocks to legal risk to sharpen the identification of our tests. Finally, we discuss the external validity of our findings by looking at the trading behavior of investors whose accusations get dismissed by Federal courts.

6.1 Regression Specification and Baseline Results

We begin by analyzing the tradeoff between information risk and legal risk as the main drivers of strategic trading decisions, *Splitting* and *Intensity*. Specifically, we estimate the following regression model:

$$\text{Strategic Decision}_i = a + b \times \text{Information Risk}_i + \text{c} \times \text{Legal Risk}_i + d \times \text{ Controls}_i \ + e \times Court_i + \text{error}_i. \tag{11}$$

The group of control variables includes empirical proxies of the economic drivers described in Section 5. Specifically, we include *Strength*, *Realized Volatility*, *Volume Vol*, and *Tipper Insider*. We also control for firm size using the natural logarithm of market capitalization, *Ln (mkt. cap)*. Finally, in all specifications we include an indicator variable (unreported) for cases with zero *Splitting* value. By doing that, we explore the intensive margin dimension of the variable. In column (2), we additionally include *Age*, *Gender*, and *Expert*. Finally, in column (3), we also include court-fixed effects. In all regressions, we cluster standard errors at the firm level.

Table IX shows the baseline results. In columns (1)-(3), we report the results for *Splitting* estimated using Tobit model. In column (1), we include *Information Risk* as the main independent variable. Consistent with our theoretical prediction, we find that the coefficient of *Information Risk* is negative and statistically significant at the 1% level of significance. The effect is also economically large, and equivalent to 40% of the standard deviation movement in *Splitting* as a result of a standard deviation change in *Information Risk*. We also find that the decision to split trades is negatively related to *Strength*, consistent with our model, and whether an insider is involved in the trade (*Tipper Insider*).

Traders also reduce their order splitting when realized variance is high. In general, there is less splitting for large stocks, consistent with the notion of lower price impact of trades for large stocks, but the coefficient is not statistically significant.

In column (2), we additionally include *Penalty* as a proxy for legal risk. In the specification, the coefficient of b still retains its negative sign and statistical significance and the coefficient of c is positive and statistically significant at the 5% level. Both effects are consistent with the hypothesis that, in their decision to split trades, traders weigh information risk against legal risk. Like that of *Information Risk*, the effect of *Penalty* is also economically significant and equivalent to a 10% movement in terms of standard deviation of *Splitting*. In column (3), we additionally include individual traders' characteristics. Both b and c remain statistically and economically significant. Further, the coefficients of *Gender* and *Expert* are both negative and statistically significant, consistent with the view that male traders are more confident in their ability to invest. The coefficient of *Age* is positive but statistically insignificant.

In columns (4)-(6), we report the results for *Intensity* estimated using OLS. The sequence of tests mimics that for *Splitting*. We find that information risk enters with the positive loading, per our hypothesis, but the effect is statistically insignificant. The coefficient of legal risk is positive as well, which seems counter to our modeling prediction, but the effect is more difficult to interpret since the size of penalty is also a function of the trade size, in general. Consistent with our theoretical predictions, we also find that traders trade less aggressively when tipper is an insider. They also trade less aggressively when the private signal is more powerful, possibly to reveal less information to the market. Finally, the intensity measure displays lower values for large market caps, which is consistent with traders facing some form of capital constraint. All other coefficients are statistically insignificant.Overall, the results from the baseline regression indicate that in their decision to split trades, insiders trade off information risk against legal risk, in the way that is consistent with our model's predictions. At the same time, the decision about trading quantity is less influenced by the two forces.

One possible concern with the interpretation of our findings is our measurement of legal risk, which may not fully capture the ex-ante decision making since we look at ex-post penalties. We believe that the effect is unlikely biased due to omitted variables. First, our regressions, in columns (3) and (6) include court fixed effects, which allows us to control for any unspecified time-invariant characteristics of courts that determine penalties and have an impact on trading strategies as well.

Second, if omitted variables drive the variation in trading strategies they should also affect the size of penalties. We test this claim formally by estimating the regression model in which we predict the size penalty, *Penalty*, and the probability of an insider going to jail, *Jail*, using various characteristics of insider trading environment. We present the results in Table X.

The results indicate that insiders' penalties are not systematically related to most of the typical predictors of insider trading activity. Hence, omitted trading characteristics are unlikely to drive the coefficient of *Penalty*. Nevertheless, unobserved omitted variables may drive the variation in penalties and trade. For that reason, we further strengthen our identification using a number of shocks to enforcement. We discuss these results in the next section.

6.2 Do Insiders Respond to Changes in Legal Risk?

Legal risk depends on two factors: the factor related to the size of penalty, conditional on the state of being caught, and the factor related to the subjective probability of being caught. In this section, we evaluate the impact of each factor separately on insiders' trading strategies.

6.2.1 Evidence from the Dodd-Frank Act

On July 21, 2010, President Obama signed into law the Dodd-Frank Wall Street Reform and Consumer Protection Act ("Dodd-Frank Act"). This landmark legislation contains a number of measures that significantly expand the enforcement authority of the SEC and strengthen its oversight and regulatory authority over the securities markets. The general objective was to create stronger deterrence to insider trading by focusing on more prominent cases and assigning larger penalties. To this end, the government increased the budget of SEC and related government agencies (see, Del Guercio et al. (2017)). As an example, in 2010, the SEC instituted the Whistleblower Reward Program that awards potential tippees with compensation related to the profits resulting from a given insider trading case. As a result of these combined activities, one could observe a significant increase in the strength of the prosecution. As an example, in 2011, the SEC has brought to court one of the largest insider trading cases ever, that of Raj Rajaratnam of the Galleon fund. At the same time, the prosecution has witnessed a major setback in 2014 with the ruling of two hedge fund managers: Todd Newman and Anthony Chiasson. Both were brought to trial but subsequently released. The Supreme Court judges' view was that no offense could be committed unless a corporate insider (the tipper) received money or valuable property (jewelry or a briefcase full of cash, say) in exchange for leaking information to the person who traded on it (the tippee). Because the defendants were several layers removed from the original leaks, it was hard to be certain that they knew that the information had been wrongfully procured. Since the government had not proved that such payments were made to the original tippers, the cases had to be dismissed. This narrow interpretation of the law torpedoed a number of insider-trading prosecutions across the Second Circuit, which happens to be where most such cases are brought.[4] The consequences of the ruling were immediate and dire. U.S. Attorney Preet Bharara,

[4]The Second Circuit includes New York and Connecticut, where many hedge funds are based.

of Manhattan, was forced to dismiss nearly a dozen of the hundred and eleven insider-trading cases that he had brought since assuming his post, in August, 2009. Among the cases that were abandoned were several in which he had already obtained guilty pleas. Many argued that 2014 reduced significantly the expected legal risk associated with insider trading.

To illustrate the developments more formally, in Figure 8, we present the time-series evolution in the number of cases resulting in jail sentence and the size of assigned penalties. One can observe a significant upward shift in the number of cases resulting in jail and the size of assigned penalties, starting in 2011. [5] At the same time, we observe a sudden drop in penalties in 2014, consistent with the ruling of Newman. Overall, the Dodd-Frank effect is economically large and lends itself as a shock to enforcement strength. Given the evidence, we believe that the period 2011-2013 signifies a strong shift in expected penalties due to insider trading. In the next set of tests, we investigate the strategic response of insider traders to the enforcement shock. Following Becker (1968), we pose that

$$\mathbb{P}_{t_{\text{info}}}\left(\text{Act on Info}_{ji}\right) = \text{Probit}\left(\mathbb{E}_{t_{\text{info}}}\left(\text{trade profit}_j\right) - \mathbb{E}_{t_{\text{info}}}\left(\text{penalty cost}_j\right), \xi_i\right)$$

where $\mathbb{P}_{t_{\text{info}}}$ reflect that probabilities are conditional to the information set available at the time of receiving the tip and ξ_i reflects personal characteristics of trader i that includes risk aversion, age, and firm position. If insider traders internalize the presence of stronger enforcement in their trading decisions, they will take decisions that lower the probability of being caught by the regulator. Hence, we should expect insiders to split their trades

[5]As an example, record-breaking criminal and civil penalties were handed out to insider trading defendants in 2011. On the criminal side, federal courts sentenced a record number of insider trading defendants and sentenced a higher proportion to prison than in any prior year. The large majority of sentences, however, remained below the Sentencing Guidelines (the "Guidelines") range and below the recommendations of prosecutors. On the civil side, the SEC settled or secured judgment in almost twice the number of insider trading cases it had brought to final judgment the year before.

more and trade less intensely. Thus, we predict that greater enforcement strength should increase the value of *Splitting* and reduce the value of *Intensity*.

To test these predictions, we estimate the following regression model:

$$\begin{aligned}\text{StrategicDecision}_i = a + b{\times}\text{Enforcement}_i{+}c{\times}\text{InformationType}_i{+}d{\times}\text{Trader}_i{+}e{\times}\ \text{Controls}_i \\ + f \times Court_i + \text{error}_i. \qquad (12)\end{aligned}$$

Enforcement is an indicator variable equal to one for insider trades taking place in the period 2011-2013, and zero for the trades in the period 2008-2010. We choose the latter period as a control period largely by symmetry and to reduce the possibility that other shocks affect our results (if the period is too long) or that the results are too noisy (if the period is too short). All regression models mimic those estimated before and we further include the set of control variables. Notably, we also include court fixed effects to control for the possibility that the legal jurisdiction may drive the strength of penalties assessed to insiders. Hence, our coefficients are driven more by the demand side rather than supply side of penalty.

We report the results in Table XI. We find the results are consistent with our hypothesis. *Enforcement* is positively related to related to *Splitting* and negatively related to *Intensity*. Both effects are statistically significant, at least at the 10% level of significance.

6.2.2 Enforcement Shocks: Cross-sectional Evidence

The perception of legal risk may vary across insider traders, depending on the traders' type and the judicial environment. In this section, we study the differences in insider trading depending on two dimensions: traders' sophistication and the strength of the prosecution.

Evidence from Whistleblower Reward Program One dimension along which we can evaluate the effect of enforcement risk is investor sophistication. On the one hand, experienced traders may be better at camouflaging their trades and thus respond more to increasing enforcement. On the other hand, knowing that they are better traders, sophisticated investors may act more boldly. In this section, we empirically evaluate this hypothesis using the specification in (1).

We use the sample of traders identified using the Whistleblower Reward Program implemented in 2011 to select experienced traders. The underlying idea of the program is to reward whistleblowers for providing *original information* directly to the SEC or related agencies. The program defines original information as one that is 1) derived from the independent knowledge or analysis of a whistleblower, 2) not known to the SEC from any other sources, and 3) not exclusively derived from an allegation made in a judicial or administrative hearing, governmental report, hearing, audit, or investigation or from the news media. Since such traders are difficult to identify it seems plausible they are more sophisticated than other traders.

Our sample thus includes 187 different cases, 47 of which were investigated through the program and 140 of which do not have a precise source of investigation (could be the result of SEC analyses or based on independent tips). In Table XII, we summarize various trading characteristics for the two types of cases. We note that the two sets of cases are not very different from each other along most of the trading dimensions. The only notable difference is that the Whistleblower Reward Program cases involve, on average, companies with greater market capitalization. Hence, it seems that the source of investigation does not seem to introduce a particular bias in terms of trading behavior.

Subsequently, we assess if more sophisticated traders respond differently to changes in enforcement environment following the adoption of Dodd-Frank Act. Formally, we estimate the following regression model:

$$\text{StrategicDecision}_i = a + b \times \text{ Controls}_i + c \times \text{Enforcement}_i + \qquad (13)$$
$$+ d \times WB + e \times Enforcement \times WB + f \times Court_i + \text{error}_i.$$

where *WB* is an indicator variable equal to one for traders identified using the WRP, and zero otherwise. *Enforcement* is defined as before. The coefficient of interest is e, which measures the relative sensitivity of two groups of traders to enforcement shocks in terms of their trading strategies. We present the results in Table XIII. In column (1), we show the results for *Splitting*. We find that sophisticated investors are more likely to split their trades when faced with greater enforcement. The coefficient of the interaction term is positive and statistically significant at the 1% level of significance. In column (2), we present the results for *Intensity*. The coefficient of the interaction term is statistically insignificant which suggests that the trading intensity of the two groups of investors does not react differently to enforcement shock. Overall, our results broadly suggest that sophisticated traders may be more responsive to increasing enforcement strength.

Evidence from the Southern District of New York Another dimension to evaluate the importance of legal risk is using the cross-sectional variation in the strength of enforcement. We take advantage of the anecdotal evidence on insider trading prosecutions, which suggests that Southern District of New York (SDNY) has become one of the toughest places to face trial under the reign of Preet Bharara, the U.S. Attorney of the District. Bharara has been appointed to the office in 2009 and has continued leading the District until 2017. As a U.S. Attorney, Bharara earned a reputation of a "crusader" prosecutor. However, the Newman ruling of 2014 has weakened his power substantially. In this regard, we argue that potential illegal traders faced particularly strong enforcement during the early times of his employment: 2009-2013.

We use this period as a shock to legal risk, and compare the insider trading strategies for traders subjected to SDNY jurisdiction to strategies of traders investigated by other offices during Bharara's time in office and periods just around the time when he carried most power. If Bharara indeed had a strong impact on the prosecution one would expect that insiders located in SDNY should become more careful with their trades during this time. Formally, we estimate the following regression model:

$$\begin{aligned}\text{StrategicDecision}_i = a + b \times SDNY_i + c \times PB + d \times SDNY \times PB \\ + e \times \ Controls_i + f \times Court_i + \text{error}_i.\end{aligned} \tag{14}$$

where *SDNY* is an indicator variable equal to one if the insider case is subject to prosecution by SDNY, and zero otherwise. *PB* is an indicator variable equal to one for the period 2009-2013, and zero for the period 2006-2008 and 2014-2015[6]. All regressions include the most comprehensive set of controls. We exclude the constant from each regression to show the direct effect of *SDNY* in our regressions. Our coefficient of interest is *d*. We present the results in columns (3) and (4) of Table XIII.

In column (3), we present the results for *Splitting.* We find no strong evidence that traders split their trades more around the time of Bharata's reign. The coefficient of d is small and statistically insignificant. In column (4), we present the results for *Intensity.* We find a strong negative effect of SDNY on trading intensity in times of Bharata's lead. The effect is economically large. Trades submitted by traders in SDNY jurisdiction are relatively less intensive in terms of their trading volume by about 20% of the standard deviation of the intensity variable.

[6]We choose this period to match the length of the ultimate power of Bharata. Because his power has weakened post 2013 we use the two years after 2013 and three years before 2009. The results are very similar if we use the five-year period of 2004-2008, instead. The results also weaken somewhat if we used the entire period of 2009-2015 as a treatment period, which is consistent with our view on his changing power to push through insider cases.

Overall, our results indicate the importance of both traders' sophistication and power of the court, which carries the investigation, for understanding the dynamics of trading strategies.

6.2.3 Evidence from Internet Search Trends

The second dimension of enforcement risk is the traders' subjective probability of legal enforcement. If the probability goes up then insiders are likely to behave more conservatively. Our model predicts that with a higher probability insider traders are more likely to split their trades and reduce the intensity of their trades. In this section, we empirically assess this hypothesis. We identify changes in the enforcement probability with periods of increased media attention with regard to insider trading. Specifically, we measure media attention using Google trends tracking the phrase "insider trading". We present the monthly variation in the attention intensity in Figure 9. The data with Google trends are available for the period 2004-2015.

Since the intensity of trends is likely not linear in the probability of legal enforcement, we specify a cut-off level, equal to the 80th percentile of the unconditional distribution of the intensity measure, as a proxy for the subjective probability of enforcement. Specifically, we define an indicator variable *Insider Trend* equal to one if the trend is above the cutoff level and zero, otherwise. Subsequently, we test the hypothesis for the effect of attention on strategic trading. Formally, we estimate the following regression model:

$$\begin{aligned}\text{StrategicTrading}_{it} = a{+}\text{b}{\times}\text{InformationTrend}_t{+}c{\times}\ \text{Controls}_{it} + \\ d_{year} + \text{error}_{it}.\end{aligned} \tag{15}$$

We present the results in Table XIV. In columns (1) and (2), we present the results for *Splitting.* We find that the coefficient of the trend is positive, but statistically insignificant, in specifications with and without year fixed effects. The result indicates that traders are more likely to split their trades in periods of higher perceived enforcement risk. Similarly, in columns (3) and (4), we report the results for *Intensity.* We find that the coefficient of *Trend* is negative and statistically significant at the 10% level of significance both in specification with and without year fixed effects.

6.2.4 Legal Risk and Signal Choices

Given that the increased legal risk increases the expected cost of insider trading, one would expect that, everything else being equal, the expected profit from insider trading falls. Consequently, insiders may react by concentrating only on information signals of high value. If this is the case, one would expect the average quality of signals obtained by insiders following the enforcement shocks to improve.

We test this hypothesis by comparing the average signal strength for the sample of trades originated in times with high strength of enforcement to that for the sample of trades originated in times with low strength of enforcement. Similarly, we perform a similar comparison for the sample of trades with high perceived risk of being caught. Formally, we conjecture that $\delta > 0$ in the following specification

$$\mathbb{E}_{t_{\text{info}}}\left(\text{trade profit}_j\right) = P_j + \delta \times \mathbb{I}\left\{t > T_{ER}\right\} + \text{error}$$

We report the results in Table XV. Consistent with our hypothesis, we find that signal strength is significantly higher in times of high legal risk, both in terms of the size of applied penalties and the probability of insiders being caught. The difference is both economically and statistically highly significant.

Overall, the evidence supports the notion that insider traders internalize the expected increase in the probability of crime detection when they decide their trades.

6.3 External Validity

The empirical evidence on the importance of legal risk exploits the variation on the intensive margin of legal risk since our sample is largely made of investors who are convicted for illegal trading. A natural question to ask is whether the results would extend to an extensive margin in which we could observe informed investors who are not facing legal risk. The empirical literature has found it difficult to understand such cases since information sets are empirically unobservable. In this paper, we are resorting to a unique opportunity in which we can precisely identify informed investors who are unlikely acting with an impression of illegal conduct.

Our sample is based on a subset of cases that have been brought to court by the SEC but eventually have been ruled out as illegal due to lack of supporting evidence. Our data include 19 such cases for a total of 121 trades. We use this sample to exploit the extensive margin of legal risk in our sample. To the extent that such traders do not act with the notion of illegality our model predicts that they should be less likely to split their trades and be more likely to trade more aggressively. We compare the trading strategies of these unconvinced traders to a sample of traders who are convicted but otherwise similar to our traders. Since the unconditional sample of convicted traders includes a very diverse set of traders, some of whom face large penalty, we focus on a subset of traders with minimal levels of penalties. Specifically, we impose the maximum level of penalty of \$140,000, which is the 25th percentile of the unconditional penalty distribution. These cutoff results in 190 trades coming from 38 different cases.

Formally, we estimate the following regression mode

$$\text{StrategicTrading}_{it} = a + \text{b} \times \text{Dismissed}_t + c \times \text{ Controls}_{it} + \text{error}_{it}. \tag{16}$$

where *Dismissed* is an indicator variable equal to one for trades that are performed by traders for whom the Federal court found no evidence of illegal behavior, and zero for trades performed by traders convicted of illegal insider trading and facing small penalties. All other controls are the same as before. We report the results in Table XVI.

We find the results that largely support our intensive margin findings. Traders who do not face legal risk are less likely to split their trades. They also trade more aggressively relative to traders who face legal risk in their trading decisions. More broadly, the fact that our results naturally extend into an extensive margin suggests that the same incentives that govern the decisions of illegal insiders also affect the behavior of legal informed investors.

7 A Brief Discussion of Implications

We analyze the optimal trading decisions of insider traders facing information risk and enforcement risk due to legal misconduct. Our research has several important implications. We provide in this section a short discussion of implications and suggest directions for future work.

Strategic use of private information Theoretically, we show that smaller information risk increases incentives to split trades and trade larger quantities, while greater legal risk increases the possibility of splitting trades and trading less aggressively. We show empirically that a large fraction of investors investigated by SEC behaves in line with this tradeoff, yet a relatively large group does not split trades at all. Further, we show that the value of information, its precision, the volatility of asset prices, and the amount of noise trading play a crucial role in designing trading strategies. We also find that individual characteristics, such as expertise and age play an important role in trading decisions. Using exogenous shocks to legal enforcement we show that investors behave more conservatively and trade assets more carefully. In light of that, insiders trade choose signals that are, on average, stronger, which suggests insider trading enforcement may indeed hamper stock price informativeness for a given firm.

The informativeness of asset prices By affecting trading strategies, the enforcement environment is likely to affect the process of price discovery and the overall informativeness of asset markets. Do insiders fully reveal the information contained in private signals? Do they reveal more or less information than other informed traders such as activist investors and hedge fund managers? Comparing the price impact of different informed traders is besides the scope of this paper. We can, however, gain perspective by zooming in on the horizon between the first day an insider trades and the day immediately after the public release of information. For that, we consider a nonparametric time scale by dividing a given insider investment horizon into ten subperiods of equal length and calculate the cumulative stock returns over the investment horizon for each trade and average them across all cases, as displayed in Figure 10. The red line represents the return path observed in the data. The average price increase over the trading horizon is about 35%. Interestingly, the total price impact up to the point of the public announcement is only 20%, that is, 60% of the total. Relative to a counterfactual risk-neutral informed trader that would smooth information transmission continuously over time in the spirit of Kyle (1985), as represented by the blue line in the figure, insider traders impound significantly less information. Moreover, a relatively large proportion of the price impact occurs during the last 10% of the available trading horizon. Further studying the determinants of the extent and speed of information aggregation of insider traders is a promising direction for future work that can clarify their impact on important market performance dimensions such as efficiency and capital allocation.

Regulatory actions Given that our findings suggest that many insiders' trading strategies are sensitive to legal enforcement risk, the resulting conservatism in trading may, in turn, make it more difficult for the regulators to detect illegal misconduct. In light of the growing sophistication of insider trading strategies, our paper rationalizes the use of alternative detection mechanisms, those which are independent of the dynamics of trading strategies, such as the Whistleblower Reward Program introduced with the Dodd-Frank Act.

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Proof of Proposition 1

We verify that the system of equations (3) to (7) is a unique linear equilibrium by following the three-step strategy in the proof of Theorem 2 in Kyle (1985). The key differences with the latter are found in the first step.

1) Trading strategy. First, we use backward induction to find optimal informed trades based on the conjectured pricing rule. Consider the informed trader problem at time $t = 2$. Disregarding second order terms, $q(x_1)q(x_2) \approx 0$, the expected penalty can be written as $K(x_1, x_2) = \frac{c}{a^2}\left(x_1^2 + x_2^2 + 2x_1x_2\right)$. Substituting K in equation (2), and re-arranging, the objective function can be written as:

$$\max_{x_2 \in \mathbb{R}} \left(v - p_1 - \frac{2c}{a^2}x_1\right) x_2 - \gamma_2 x_2^2,$$

where $\gamma_t := \left(\lambda_t + \frac{c}{a^2}\right)$. Assuming that a is large enough so that x_2 lies in $[-a, a]$, an interior solution must satisfy the following first-order condition:

$$x_2^* = \frac{\left(v - p_1 - \frac{2c}{a^2}x_1\right)}{2\gamma_2} \tag{17}$$

$$= \frac{1}{2\gamma_2}\left(v\left(1 - \frac{2c}{a^2}\beta_1\right) - p_1 - \frac{2c}{a^2}\alpha_1\right). \tag{18}$$

Note that, unlike in the traditional Kyle model, the optimal x_2 depends on the value of x_1 because enforcement risk creates a new intertemporal dependency. Satisfying the second order condition requires $\gamma_2 > 0$.

Using (17), the value function V_2 can be expressed as

$$V_2 = \mathbb{E}\left[x_2^*\left(v - p_1 - \lambda_2(u_2 + x_2^*)\right) \middle| \, v, p_1\right] - \frac{c}{a^2}\left(x_1^2 + x_2^{*2} + 2x_1x_2^*\right),$$

$$\Rightarrow V_2 = \frac{\left(v - p_1 - \frac{2c}{a^2}x_1\right)^2}{4\gamma_2} - \frac{c}{a^2}x_1^2. \tag{19}$$

Consider now the insider problem in period $t = 1$. Using equation (19) we can re-express the program in equation (1) as

$$\max_{x_1 \in \mathbb{R}} \mathbb{E}\left[(v - p_0 - \lambda_1 y_1)\, x_1 + \frac{(1-\rho)}{4\gamma_2} \left(v - p_1 - \frac{2c}{a^2} x_1 \right)^2 |v \right] - \frac{c}{a^2} x_1^2. \tag{20}$$

Expanding the term $\left(v - p_1 - \frac{2c}{a^2} x_1\right)^2$, evaluating the expectation, and dropping constant terms, one can write (20) as

$$\max_{x_1 \in \mathbb{R}} \left\{ (v - p_0)\, x_1 \left(1 - 2\theta \left(\gamma_1 + \frac{c}{a^2} \right) \right) + \left(\theta \left(\gamma_1 + \frac{c}{a^2} \right)^2 - \gamma_1 \right) x_1^2 \right\},$$

where $\theta_2 = \frac{(1-\rho)}{4\gamma_2}$. The first-order condition is

$$(v - p_0) \left(1 - 2\theta \left(\gamma_1 + \frac{c}{a^2} \right) \right) = 2 \left(\gamma_1 - \theta \left(\gamma_1 + \frac{c}{a^2} \right)^2 \right) x_1,$$

implying that

$$x_1^* = \frac{(v - p_0) \left(1 - 2 \left(\gamma_1 + \frac{c}{a^2} \right) \theta \right)}{2 \left(\gamma_1 - \left(\gamma_1 + \frac{c}{a^2} \right)^2 \theta \right)}. \tag{21}$$

Therefore, $\alpha_1 = -\beta_1 p_0$ and

$$\beta_1 = \frac{\left(1 - 2 \left(\gamma_1 + \frac{c}{a^2} \right) \theta \right)}{2 \left(\gamma_1 - \left(\gamma_1 + \frac{c}{a^2} \right)^2 \theta \right)}.$$

The second-order condition can be expressed as $\lambda_1 > \theta \left(\lambda_1 + \frac{2c}{a^2} \right)^2 - \frac{c}{a^2}$.

To find α_2 and β_2, we combine equations (21) and (17) to obtain

$$x_2^* = \frac{1}{2\gamma_2} \left(v \left(1 - \frac{2c}{a^2} \beta_1 \right) - p_1 + \frac{2c}{a^2} \beta_1 p_0 \right).$$

Therefore,

$$\beta_2 = \frac{1}{2\gamma_2}\left(1 - \frac{2c}{a^2}\beta_1\right),$$
$$\alpha_2 = \frac{1}{2\gamma_2}\left(\frac{2c}{a^2}\beta_1 p_0 - p_1\right).$$

2) Learning and prices. Based on the insider strategy and using a market efficiency condition, we address the market maker learning problem and derive the pricing rule. In period $t = 1$, informational efficiency requires that the asset satisfies $p_1 - p_0 = \mathbb{E}\left[v - p_0 | \alpha_1 + \beta_1 v + u_1 = y_1\right]$. Given the normal distribution assumption for v and u_1, the projection theorem implies that λ_1 must be given by (5). In period $t = 2$, based on the signals y_1 and y_2, informational efficiency requires that $p_2 - p_1 = \mathbb{E}\left[v - p_1 | \alpha_2 + \beta_2 v + u_2 = y_2\right]$. Then,

$$\mathbb{E}\left(v - p_1 | y_1, y_2\right) = \frac{\beta_2 \sigma^2_{v|y_1}}{\beta_2^2 \sigma^2_{v|y_1} + \sigma_u^2} y_2, \tag{22}$$

Using equation (22) and $\sigma^2_{v|y_1} = \frac{\sigma_v^2 \sigma_u^2}{\beta_1^2 \sigma_v^2 + \sigma_u^2}$ yields $\lambda_2 = \frac{\beta_2 \sigma_v^2}{\left(\beta_2^2 + \beta_1^2\right)\sigma_v^2 + \sigma_u^2}$.

3) Having characterized the linear structure of trading strategies and pricing rule in 1) and 2), the existence of a unique way to iterate the system backward follows from the arguments given in Kyle (1985). ■

Figure 3. Chronology of the Elan/Wyeth Insider Trading Scandal

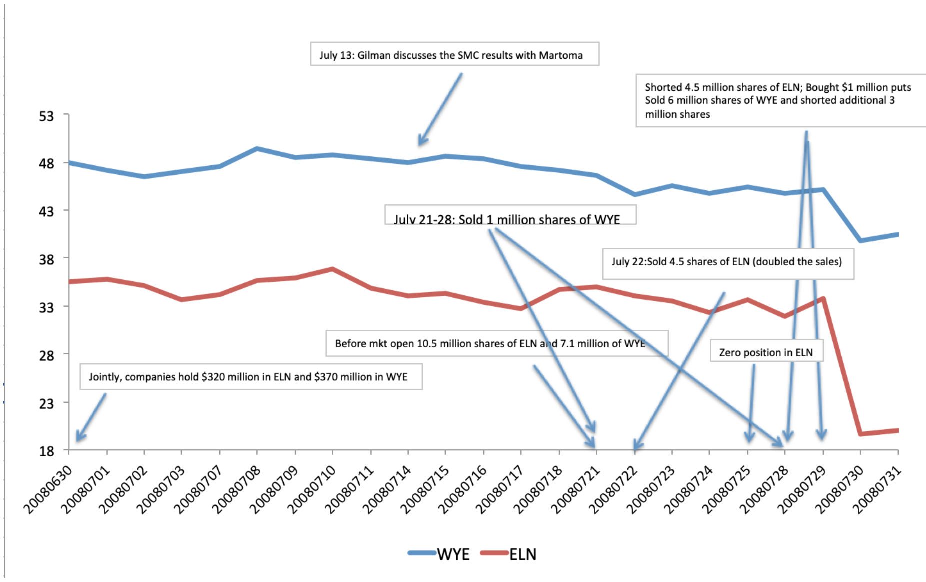


Figure 4. Stock Prices of Elan and Wyeth

Figure 5. Equilibrium: Information and Enforcement Risks

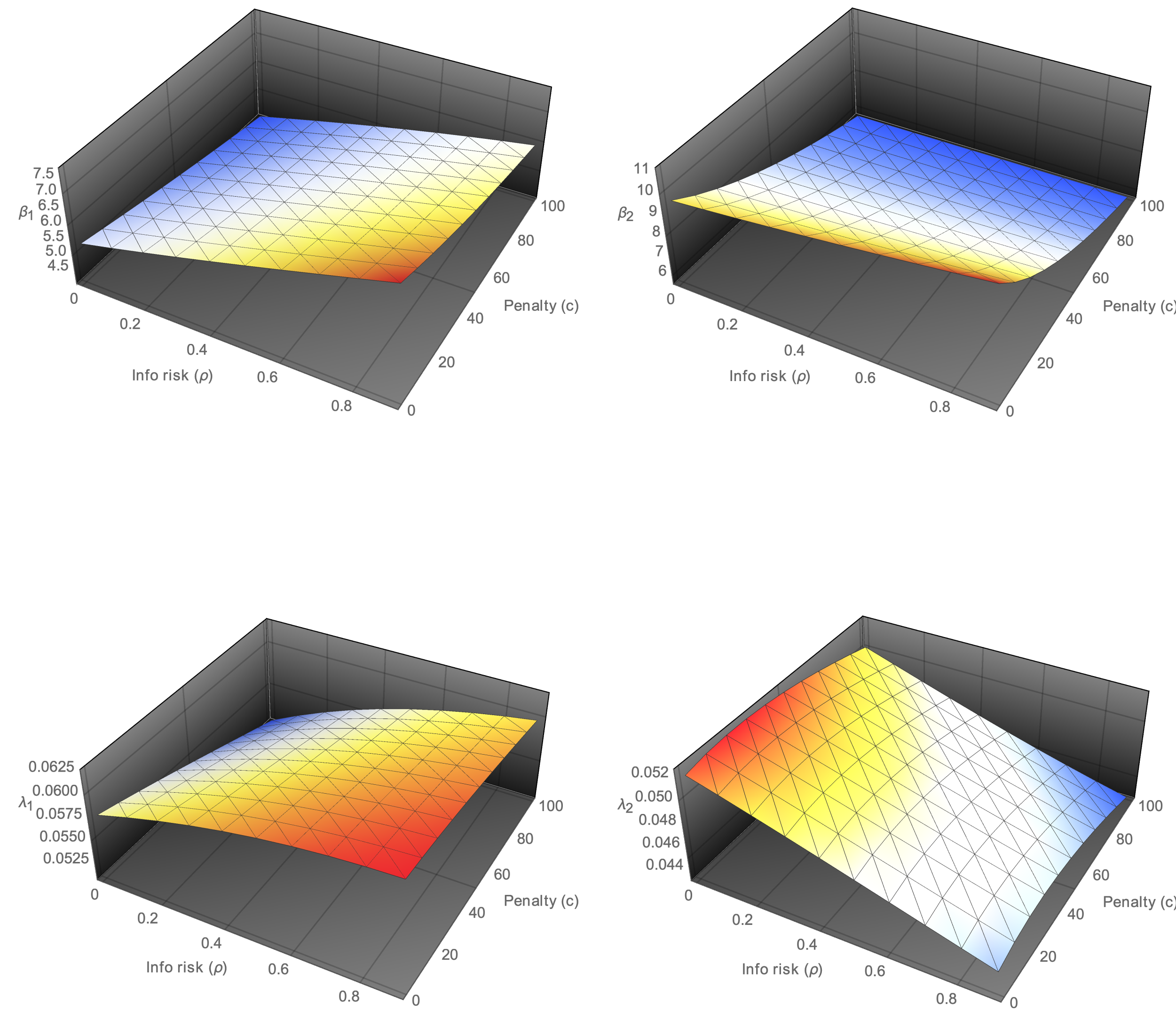


Figure 6. Timeline of Events

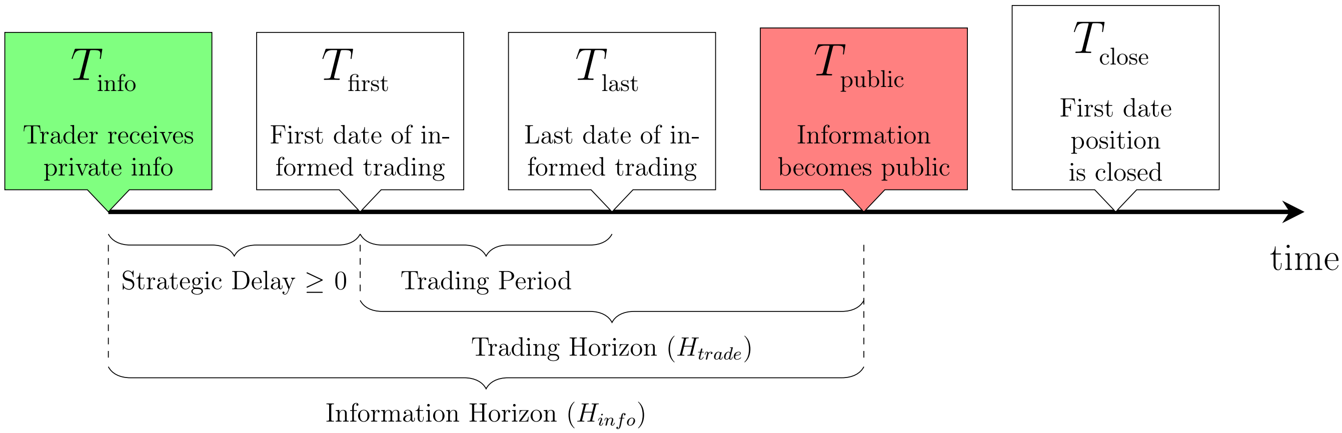
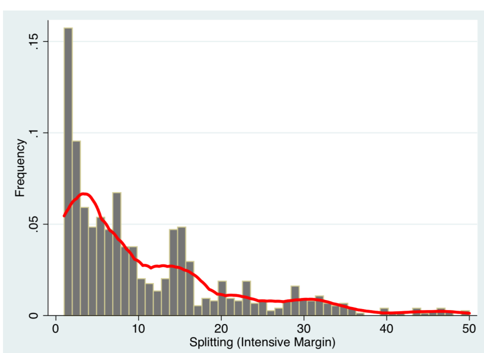
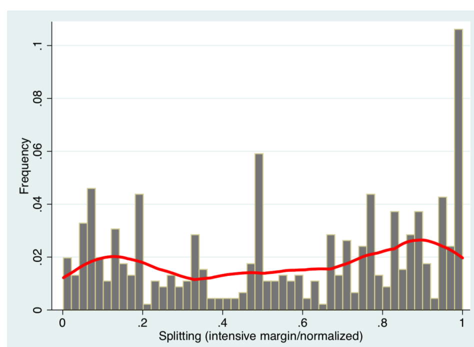


Figure 7. Empirical Distributions of Splitting and Intensity

(a) Splitting



(b) Splitting -Normalized



(c) Asset Intensity

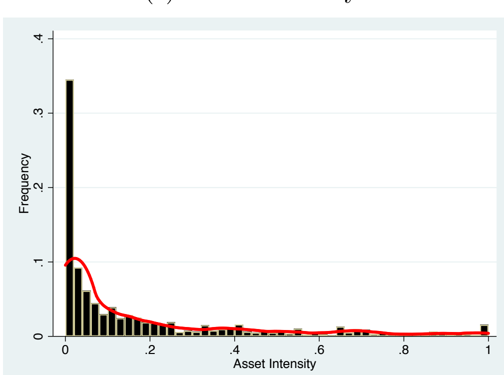


Figure 8. Penalties and Jail Sentences over Time

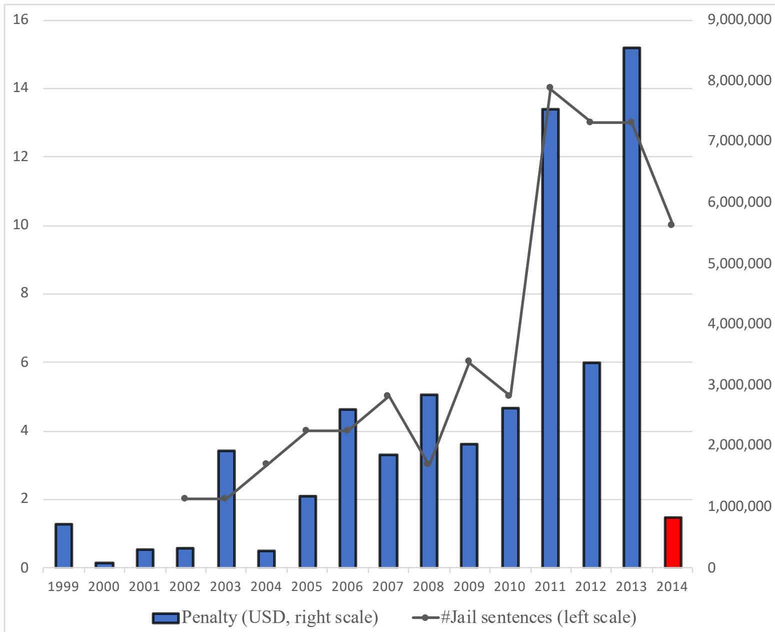
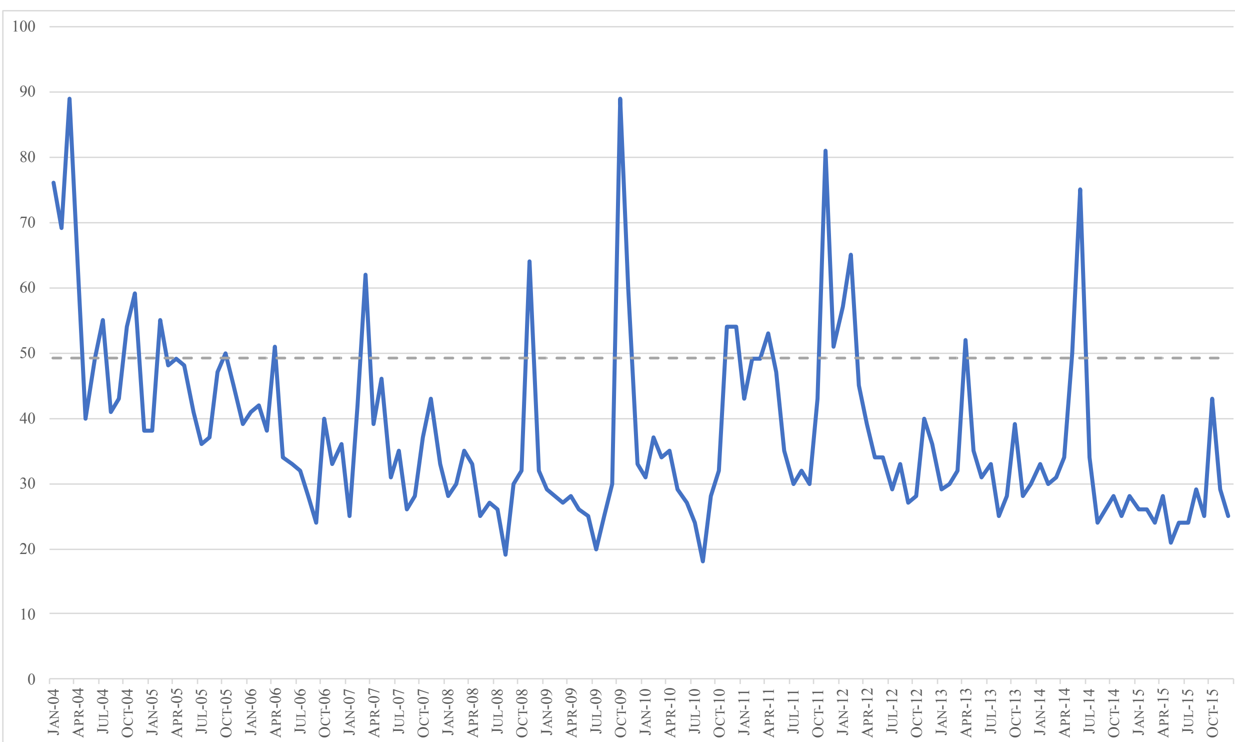


Figure 9. Internet Search Intensity for 'Insider Trading': January 2004 to December 2015



Source: Google Trends.

Figure 10. Trades and Information Transmission

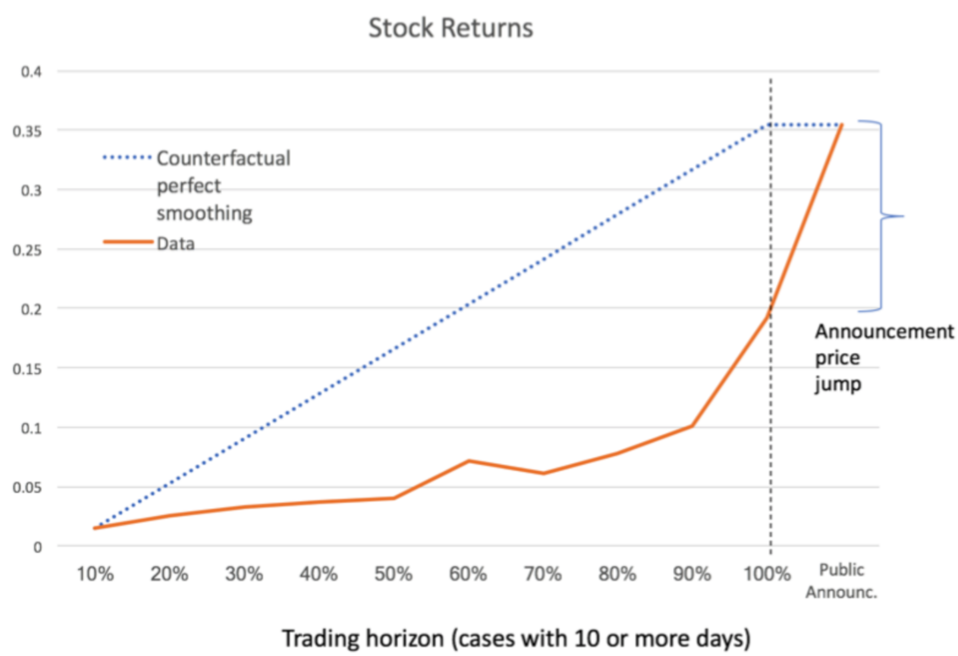


Table I
Summary Statistics: Events and Trading Instruments

We classify the sample of SEC insider trading cases based on event type (M&A, earnings announcements, general business events, shares offerings, and dividend changes), information sign (good news/bad news), and trading instrument (stocks, options, ADS, and bonds).

Event type	Number of Firm/Cases	Percentage of Firm/Cases
Mergers & Acquisitions	360	55.90
Earnings announcements	97	15.06
General business events (patents, trials)	69	10.71
Shares offerings & tenders	52	8.07
Dividend changes & buybacks	14	2.17
Other (restatements/fraud/manipulation)	52	8.07
Information sign		
Good news	664	71.86
Bad news	260	28.14
Trading instrument		
Stocks	3,392	67.06
Options	1,610	31.83
ADS	44	0.87
Bonds	12	0.24

Table II
Summary Statistics: Trades, Traders, and Tippers

We classify the sample of SEC insider trading cases. Firms per case is the number of distinct companies reported in a given case; traders per case is the number of distinct traders involved in a given case; trades per trader is the number of trades executed by insider in all cases he/she appears in the data; trades per firm is the number of trades executed by all insiders trading a given firm's shares; trader finance background is the percentage of insiders who have finance job; trader top executive is the percentage of insiders who hold executive positions in their respective companies; reported profit is the dollar profit realized on a given trade (in '000).

Characteristic	mean	median	st. dev.	min	max
Firms per case (N=840)	2.07	1	3.15	1	26
Traders per case (N=1079)	5.71	4	5.21	1	25
Trades per trader	20.22	10	24.32	1	97
Trades per firm	18.53	13	21.24	1	126
Trader age	47.16	46	11.78	22	91
Tipper age	46.26	45	11.64	25	80
Trader gender (male in %)	91.89	-	-	-	-
Trader finance background (in %)	60.06	-	-	-	-
Trader top executive (in %)	29.97	-	-	-	-
Reported profit ($1,000s)	1013.64	90.00	7926.79	4.0	27500

TABLE III
Trading Sophistication: Summary Statistics

Percentage of zero values	Splitting
All information events	69.6
Earnings Announcements	58.2
Mergers & Acquisitions	74.5
Information sign	
Good news	71.9
Bad news	59.5
Trading Instrument	
Stocks	70.4
Options	69.0

TABLE IV
Measuring the Information Content of Trades

The return is based on stock price data and is computed from the open price on the insider trading day to the open price on the day following the public disclosure date. The returns are split according to positive and negative news. The aggregate return considers the absolute value of each return. The returns for 13D filers are measured from the open price of the day 13D filers trade to the open price of the day following the public disclosure date of the trade. ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Positive	Negative	Aggregate	
	Illegal Insider Trading			13D Filers
Mean Return (%)	43.510***	-18.564***	38.271***	4.927***
	(4.199)	(2.142)	(3.389)	(0.638)
Median Return (%)	33.690***	-15.322***	29.427***	2.401***
	(2.348)	(2.545)	(2.275)	(0.173)
#Obs	2,351	696	3,055	2,628

Table V
Traders Professional Background and Corporate Positions

This table reports the numbers of insider traders in our sample distributed according to their job category. In **Panel A**, we break down the sample by the corporate function; **in Panel B**, we break down the sample according to the types of finance jobs; in Panel C, we summarize the data according to non-finance jobs. The sample period is 1995–2015.

Category	**Number**
Panel A: Corporate Function	
Employee/Low-level management	105
Mid-level management	84
Vice president	67
CEO	38
Director	32
President	32
CFO	24
Other	28
Panel B: Job in Finance	
Portfolio manager/Hedge fund	59
Broker or dealer	30
Analyst	21
Trader	16
Financial Advisor	7
Other	16
Panel C: Non-finance Jobs	
Business owner/self employed	52
Lawyer/Attorney	42
Medical doctor / dentist	24
Accountant	14
Sales	13
Engineer /IT	12
Real estate broker	11
Other	26

Table VI
Baseline Model: Summary Statistics

Splitting is the difference in number of days from the first to the last trade, scaled by the number of days from the date of information arrival until the day information is public. *Intensity* is the maximum over all traded assets of the asset-specific trade size relative to the volume in the same asset. *Strength* is the return on information recorded from the open of the day when the insider trades until the open of the day following the public disclosure of information; *Alt Strength* is the return on information recorded from the open of the day when the insider receives information until the open of the day following the public disclosure of information; *Realized Variance* is the return volatility of a traded stock; *Volume Vol* is the volatility of trading volume; *Tipper Insider* is an indicator variable equal to one if the insider is either a tipper or a trader in a case; *Expert* is an indicator variable equal one if the trader is a corporate insider and zero otherwise; *Traders* is the number of distinct traders involved in trading of a given stock and case; *Age* is the age of trader in years; *Gender* is an indicator variable if a trader is male, and zero if a trader is female; *Ln(mkt. cap)* is the natural logarithm of the firm's market capitalization.

Characteristic	mean	median	st. dev.
Splitting	0.171	0	0.316
Intensity	-0.057	-0.099	0.228
Strength ($v/p_0^T - 1$)	30.61	24.74	54.96
Alt Strength ($v/p_0^R - 1$)	34.64	27.18	76.22
Realized Volatility (σ_v)	15.15	5.24	41.91
Volume Vol. (σ_z)	134.99	54.32	255.65
Tipper Insider ($1/\sigma_e$)	24.51	0	43.02
Information Risk (π)	76.95	1	42.12
Traders	5.20	3	4.74
Expert	50.49	50.00	46.35
Ln(mkt. cap)	13.89	13.88	1.82

Table VII
Information Risk and Strategic Delay

The dependent variable is *Strategic Delay* (defined as the difference in number of days from the date when the trader receives information until the first time she trades, scaled by the number of days from the date of information arrival until the day information is public). All controls are defined in Table VI. We estimate the regression using Tobit model. ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Strategic Delay		
Information Risk (π)	-0.186***	-0.137***	-0.115**
	(0.042)	(0.052)	(0.050)
Traders	-0.003	-0.001	-0.002
	(0.004)	(0.004)	(0.005)
Strength ($v/p_0 - 1$)		-0.064**	-0.069**
		(0.028)	(0.028)
Realized Variance (σ_v^2)		0.365***	0.353***
		(0.132)	(0.132)
Volume Vol. (σ_z^2)		-0.005	-0.001
		(0.009)	(0.009)
Tipper Insider ($1/\sigma_e$)		0.007	0.005
		(0.036)	(0.040)
Ln(mkt. cap)		0.018	0.012
		(0.015)	(0.015)
Age			0.001
			(0.001)
Gender			0.074
			(0.065)
Observations	2,912	2,313	2,051

Table VIII
Insider Penalties: Summary Statistics

We classify the sample of SEC insider trading cases. Firms per case is the number of distinct companies reported in a given case; traders per case is the number of distinct traders involved in a given case; trades per trader is the number of trades executed by insider in all cases he/she appears in the data; trades per firm is the number of trades executed by all insiders trading a given firm's shares; trader finance background is the percentage of insiders who have finance job; trader top executive is the percentage of insiders who hold executive positions in their respective companies; reported profit is the dollar profit realized on a given trade (in '000).

Characteristic	mean	median	st. dev.	min	max
Trader Penalty (in $millions)	2.85	0.20	14.03	0	156.61
Total Penalty Per Case (in $millions)	11.74	1.25	32.68	0	177.21
St. Dev. of Trader Penalty Per Case (in $millions)	3.16	0.08	11.37	0	89.64
Jail (in %)	10.48	-	-	-	-
Percentage of traders in jail (per case)	10.43	0	22.24	0	100
Probation (in %)	23.55	-	-	-	-
Dropped (in %)	12.40	-	-	-	-
Most Active Courts (N=77)					
Southern District of New York	144	30.44%	-	-	-
District of Columbia	29	6.13%			
Central District of California	26	5.50%			

Table IX
Determinants of Trading Strategies

The dependent variables are *Splitting* and *Intensity*. All variables are defined in Table VI. We estimate the regression using Tobit model (in columns 1-3) and OLS (in columns 4-6). ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

		Splitting			**Intensity**	
Information Risk (π)	-0.298***	-0.230***	-0.245***	0.007	0.006	0.016
	(0.089)	(0.084)	(0.094)	(0.014)	(0.014)	(0.017)
Penalty (c)		0.002**	0.002**		0.001***	0.001***
		(0.001)	(0.001)		(0.000)	(0.000)
Strength ($v/p_0 - 1$)	-0.026	-0.040	-0.078	-0.031**	-0.030**	-0.020
	(0.057)	(0.060)	(0.084)	(0.015)	(0.015)	(0.013)
Realized Volatility (σ_v)	-0.405***	-0.387***	-0.316**	-0.010	-0.010	-0.040
	(0.114)	(0.106)	(0.125)	(0.062)	(0.062)	(0.059)
Volume Vol. (σ_z)	0.008	-0.004	0.005	0.005*	0.004	0.005
	(0.013)	(0.011)	(0.015)	(0.003)	(0.003)	(0.003)
Tipper Insider ($1/\sigma_e$)	-0.167***	-0.174***	-0.189***	-0.031**	-0.034**	-0.044***
	(0.058)	(0.057)	(0.071)	(0.015)	(0.016)	(0.017)
Ln(mkt. cap)	-0.039	-0.033	-0.041	-0.035***	-0.037***	-0.043***
	(0.026)	(0.023)	(0.027)	(0.008)	(0.008)	(0.009)
Age			0.005			-0.001
			(0.004)			(0.001)
Gender			-0.131**			-0.119
			(0.062)			(0.090)
Expert			-0.180**			0.019
			(0.076)			(0.017)
Court F.E.	No	No	Yes	No	No	Yes
Observations	2,303	1,961	1,639	3,562	3,076	2,594

Table X
Determinants of Penalties

Penalty is a dollar amount of assessed penalty (in \$million). *Jail* is an indicator variable equal to one if an insider receives a jail sentence, and zero, otherwise. All controls are defined in Table VI Court fixed effects are related to the court in which the settlement takes place. We estimate all regressions using OLS model. ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Penalty		**Jail**	
Information Risk (π)	-0.660	-1.042	-0.009	-0.073
	(4.688)	(5.273)	(0.068)	(0.061)
Strength ($v/p_0 - 1$)	-1.159	-0.712	-0.039	-0.018
	(1.187)	(1.096)	(0.045)	(0.034)
Realized Volatility (σ_v)	-1.738	-1.869	-0.114	0.037
	(7.059)	(7.991)	(0.159)	(0.151)
Volume Vol. (σ_z)	2.127	1.739	0.020	0.017
	(1.477)	(1.564)	(0.012)	(0.012)
Tipper Insider ($1/\sigma_e$)	-1.606	-0.433	-0.139**	-0.068
	(3.734)	(4.438)	(0.056)	(0.055)
Ln(mkt. cap)	2.085	2.214	0.004	0.017
	(1.373)	(1.532)	(0.016)	(0.015)
Age	0.249**	0.350**	0.002	0.000
	(0.100)	(0.147)	(0.002)	(0.002)
Gender	6.381**	9.968**	-0.007	-0.040
	(3.238)	(4.577)	(0.092)	(0.100)
Expert	3.993*	2.953	-0.010	-0.006
	(2.273)	(2.306)	(0.058)	(0.048)
Court F.E.	No	Yes	No	Yes
Observations	2,660	2,660	2,840	2,840

Table XI

Trading Strategies and the Enforcement Strength

The dependent variables are *Splitting* and *Intensity,* defined in Table IX. *Enforcement* is an indicator variable equal to one for the period 2011-2015, and zero for the period up to and including 2010. All controls are defined in Table VI. We estimate the regression using Tobit model (in columns 1 and 2) and OLS (in columns 3 and 4). ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Splitting		**Intensity**	
Enforcement	0.096**	0.086*	-0.042*	-0.054*
	(0.038)	(0.054)	(0.025)	(0.032)
Information Risk (π)	-0.281**	-0.236**	0.020	0.021
	(0.136)	(0.119)	(0.030)	(0.033)
Strength ($v/p_0 - 1$)	-0.207***	-0.203***	-0.037**	-0.038**
	(0.060)	(0.066)	(0.011)	(0.013)
Realized Volatility (σ_v)	-0.404***	-0.389***	-0.009	-0.028
	(0.101)	(0.090)	(0.033)	(0.035)
Volume Vol. (σ_z)	0.022	0.012	-0.001	-0.000
	(0.020)	(0.021)	(0.003)	(0.003)
Tipper Insider ($1/\sigma_e$)	-0.110	-0.163*	-0.012	-0.018
	(0.068)	(0.097)	(0.019)	(0.027)
Ln(mkt. cap)	-0.077	-0.074*	-0.020*	-0.025**
	(0.048)	(0.040)	(0.009)	(0.009)
Age		-0.002		-0.000
		(0.005)		(0.000)
Gender		-0.246***		-0.025
		(0.035)		(0.012)
Court F.E.	Yes	Yes	Yes	Yes
Observations	1,201	1,034	1,685	1,472

Table XII

SEC Whistleblower Cases: Summary Statistics

We consider the insider trading trades over the period 2008-2013. We define two sets of insider trading cases: WB=1 are cases that have been investigated by the SEC based on Whistleblower Reward Program; WB=0 are the cases with unknown origin of investigation sampled during the sample time period.

Characteristic/Sample	WB=1	WB=0
Number of Cases	47	140
Distance from news to trade	6.82	7.60
Distance from trade to event	15.48	17.46
Distance from first to last trade	7.59	6.85
Trades per firm	5.61	7.33
Trades per trader	8.29	9.92
Market capitalization (in Billions)	10.27	5.82
Reported profits (in Millions)	0.98	1.31

Table XIII
Trading Strategies and Enforcement Strength: Cross-Sectional Evidence

The dependent variables are *Splitting* and *Intensity,* defined in Table IX. *Enforcement* is an indicator variable equal to one for the period 2011-2013, and zero for the period 2008-2010. *WB* is an indicator variable for insider trades detected through the Whistleblower Program, and zero for trades detected through other means. *PB* is an indicator variable equal to one for the period 2009-2013, and zero for the period 2006-2008 and 2014-2015. *SDNY* is an indicator variable for traders convicted by Southern District of New York, and zero for traders convicted in other courts. All controls are defined in Table VI. We estimate the regression using Tobit model (in columns 1 and 3) and OLS (in columns 2 and 4). ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Splitting	**Intensity**	**Splitting**	**Intensity**
Enforcement	-0.200	-0.055*		
	(0.124)	(0.033)		
WB	0.171*	-0.002		
	(0.089)	(0.016)		
Enforcement*WB	1.926***	0.015		
	(0.227)	(0.031)		
SDNY			2.433**	0.395***
			(1.216)	(0.105)
PB			-0.172	-0.004
			(0.182)	(0.012)
SDNY*PB			-0.204	-0.035**
			(0.263)	(0.014)
Controls	Yes	Yes	Yes	Yes
Court F.E.	Yes	Yes	Yes	Yes
Observations	1,036	1,472	1,197	1,858

Table XIV
Trading Strategies and Perceived Legal Risk

The dependent variables are *Splitting* and *Intensity,* defined in Table IX. *Insider Trend* is an indicator variable equal to one for the period with google trends of the phrase "insider trading" greater than the 80% of the unconditional distribution, and zero otherwise. All controls are defined in Table VI. We estimate the regression using Tobit model (in columns 1 and 2) and OLS (in columns 3 and 4). ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Splitting		**Intensity**	
Insider Trend	0.041	0.035	-0.022*	-0.030*
	(0.050)	(0.054)	(0.014)	(0.017)
Year F.E.	No	Yes	No	Yes
Observations	1,201	1,034	1,685	1,472

Table XV
Signal Strength and Enforcement

Low Strength refers to cases which were originated during the period 2008-2010; and *High Strength* refers to cases which were originated during the period 2011-2013. *Low Risk* refers to cases when google trends indicator for insider trading is below 80% value of the distribution; and *High Risk* refers to cases when google trends indicator for insider trading is above 80% value of the distribution. *Strength* is defined in Table *VI*. p-values correspond to the statistical difference in means between the two periods.

Signal Strength	Low Strength	High Strength	Post-Pre (p-val.)	Low Risk	High Risk	High-Low (p-val.)
Average (%)	25.00	31.90	0.017	26.37	22.09	0.044
Observations	1,393	686	-	759	2,721	-

Table XVI
Evidence from Dismissed Cases

The dependent variables are *Splitting* and *Intensity*. *Dismissed* is an indicator variable equal to one for trades that are performed by traders for whom the Federal court found no evidence of illegal behavior, and zero for trades performed by traders convicted of illegal insider trading and facing small penalties. All control variables are defined in Table VI. We estimate the regression using Tobit model (in columns 1-2) and OLS (in columns 3-4). ***, **, * denote 1%, 5%, and 10% level of statistical significance, respectively.

	Splitting		**Intensity**	
Dismissed	-0.587**	-0.651**	0.156**	0.153*
	(0.292)	(0.278)	(0.073)	(0.078)
Information Risk (π)	-1.147***	-1.313***	-0.104**	-0.055
	(0.358)	(0.428)	(0.047)	(0.052)
Strength ($v/p_0 - 1$)	0.290	0.658**	-0.024	-0.029
	(0.275)	(0.276)	(0.093)	(0.098)
Realized Volatility (σ_v)	2.937***	-0.887	0.033	0.338
	(0.889)	(1.056)	(0.258)	(0.276)
Volume Vol. (σ_z)	0.051	0.049	-0.003	-0.003
	(0.045)	(0.030)	(0.008)	(0.009)
Tipper Insider ($1/\sigma_e$)	0.409**	0.023	-0.059	-0.049
	(0.191)	(0.158)	(0.040)	(0.042)
Ln(mkt. cap)	-0.112	-0.275***	-0.042**	-0.034
	(0.076)	(0.088)	(0.020)	(0.021)
Age		-0.028**		0.003
		(0.011)		(0.003)
Gender		-0.940***		-0.072
		(0.210)		(0.058)
Expert		0.546*		0.016
		(0.282)		(0.068)
Observations	169	149	242	228